

Making History in Linked Statistical Data: Towards a Humanities Enabled Semantic Web

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Abstract

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Chapter 1

Introduction

In recent years, we have seen the rise of structured data on the Web. The Semantic Web has been leading the revolution of extending a Web once only understandable by humans (based on the paradigm of the document) into a Web processable for machines (based on the paradigm of Linked Data and meaning). At the same time, an avid world of accountancy and transparency has been claiming more Open Government Data (OGD). As a combination of the emerging Linked Data Web and the urge of more OGV, governments and public agencies have been releasing government data as Open Data on the Web, with varying topics and several levels of structuredness.

Some of these datasets contain rich statistical data. Of these, a few are also historical data. Historical statistics, such as censuses, are a source of invaluable knowledge for social historians –historians concerned about the experiences of ordinary people in the past. The main research question of this thesis is:

Under which circumstances can historical statistical sources be modeled and published on the Web using Semantic Web standards, in such a way that their integration is increased and their analysis over time is facilitated?

Data integration is a major issue of statistical and historical sources of any kind. In statistics, it is sometimes useful to combine data from different datasets to gain insight into phenomena. Such combination requires matching statistical variables. However, deciding whether two arbitrary variables can be faithfully compared is no trivial task. In historical research, record linkage matches people across different census records or parish registers in order to reconstruct their complete story. The lack of personal identification numbers, standard spellings for names, changing family names or changing boundaries in the territory are good examples of data integration problems in this domain.

To solve these integration issues, researchers use background knowledge. However, background knowledge is not always available in statistical data, and it is very scarce in historical data. Their availability in a structured format on the Web is even more compromised. In the absence of such knowledge on the Web providing meaning of statistical historical entities, such integration is hardly realizable.

Another fundamental issue that what social historians (and historians in general) study is change over time. A flip on a trend, the subtle or sudden appearance of

some epoch-changing event, are all instances of what historians are most interested in: change.

- The flux of things is one ultimate generalization around which we must weave our philosophical system. (A. N. Whitehead, *Process and Reality*)
- I am not now That which I have been (Lord Byron)
- But now, as he stares at the relic bo tree, something shifts.
- Nothing lasts forever. A kuti is a cell. This cell is a prison. He sits in a prison, while the ones who took Chaya live and drink and whore and laugh. Nothing is permanent. This is the central teaching of the Buddha. Not a career, not an institution, not a wife, not a tree... All is change; change is the only truth. (Paolo Bacigalupi, *The Windup Girl*)

This thesis contributes on solving these two issues (namely, the integration of statistical data and the study concept drift over time) using Semantic Web standards, tools and methods. The chapters of the thesis are distributed on three different parts: the first two corresponding to one of the two issues; and an additional one studying the nature of Historiography in the Semantic Web.

- Part I. Integration of Linked Statistical Data
- Chapter 2. LSD Dimensions: Statistics over Linked Statistical Data
- Chapter 3. Linking Statistical Data with TabLinker
- (Chapter 3.1. TabCluster: Bottom-up Concept Scheme Generation in Linked Statistical Data)
- Chapter 4. Comparability of Linked Statistical Data Analysis
- Part II. Meaning Drift Over Time in Linked Historical Data
- Chapter 5. Extensional Concept Drift with Statistics
- Chapter 6. Predicting Concept Drift in Linked Data
- Chapter 7. The LOD Clairvoyant: Change of Meaning in Linked Data
- Part III. Historical Data in the Semantic Web
- Chapter 8. What is Linked Historical Data?
- Chapter 9. Semantic Technologies for Historical Research

Part I

What is Linked Historical Data?

Chapter 2

Semantic Web for the Humanities

2.1 Motivation and research questions

Humanities researchers have been interested recently in publishing and linking their datasets following Linked Data principles, in order to enhance their decentralization, openness, changeability and integration. Traditionally, the unique demands of the Humanities, their limited technical and modelling interests, and the highly contextualized nature of their source materials have kept this field distant from the Semantic Web.

We make efforts to bridge the gap. As case studies, we convert Dutch historical censuses (1795-1971) and the catalogue of publications in the Netherlands during the Golden Age (STCN, 16th century onwards) to RDF [10], we model them using standard vocabularies, and we publish them on the Web.

These datasets are messy and heterogeneous. Different dataset versions contain inconsistent structuring rules, concepts with a changing meaning over time, and multiple representation formats. Comparison, combination and longitudinal queries (e.g. evolution on the number of shoemakers in Amsterdam from 1795 to 1971) are notoriously difficult. Researchers are forced to manually rewrite data and queries, incurring in high labour costs and non repeatable practices.

Figure 1 shows data heterogeneity interacting with other indicators. Since our goal is to increase data integration, data heterogeneity has to lower, as shown by arrows and signs. Lowering data heterogeneity is no trivial task, and we identify format round-tripping, concept drift and contextual knowledge as influencing indicators that can indirectly improve data integration.

Format round-tripping. Lots of data formats are used to encode semistructured datasets. Tools for legacy conversion between these formats are required: Humanities researchers use non RDF compatible tools, and providing data in

Fig. 1: Indicators influence each other as indicated by arrows. The increase of any given indicator increases/decreases, respectively, the one influenced by it as indicated by the +/- signs. E.g., increasing format round-tripping decreases data heterogeneity.

various formats on demand is a requirement. Under this topic we will investigate, first, how to perform any plain, tabular, tree-based, graph-based or relational- based format conversion from a holistic point of view and, second, whether the original data can be retrieved after arbitrary conversions.

Concept drift. Different versions of the same dataset show that concepts change their meaning over time, especially if the time gap is wide. Although not meaning exactly the same, two time gapped instances of the same concept may preserve some degree of sameness. For example, the concept shoemaker in the 17th century (someone who makes shoes with leather) has drifted until nowadays (someone who owns a company). Mapping drifted concepts correctly is necessary to solve longitudinal queries in Humanities data.

Contextual knowledge. Humanities ontologies require dynamic concept formalizations instead of static ones, especially for contested, open-textured or ambiguous concepts. The definition of such concepts needs to be dynamically built depending on their contexts. Examples of contextual knowledge are the time when and the space where the concept occurs, subjective opinions on the concept, or domain expert statements about the concept. Multiple contradictory definitions may need to coexist in one ontology.

Concept drift and knowledge from contexts are closely related. Since the context of a concept often changes over time, a definition of concept drift based on the varying properties in contexts can be established. Despite less connected, formats often define metadata describing dataset contextual information, which needs to be appropriately modelled. We realize these phenomena are not exclusive to the Humanities, and this proposal looks further on solving longitudinal analyses in dynamic domains of any kind.

We define a general goal of providing algorithms, formalisms and tools to disambiguate, clean, prepare, normalize, transform, link and query Humanities datasets, conforming a framework for effective Humanities data publishing in the Semantic Web. Under this umbrella, our research questions are:

1. Can RDF data models faithfully represent the Humanities sources? Is an RDF-based format round-tripping framework possible?
2. How can we model concept drift? Can drifted concepts be aligned?
3. Can we infer dynamic concept definitions from explicitly formalized contexts? Can these contexts help solving concept drift?

2.2 State of the art

Work has been developed on translating RDF, spreadsheet formats and relational databases. Conversion from relational databases to RDF is covered by [7,13], and the W3C has developed a standard (R2RML) for this purpose. Some tools like D2RQ allow accessing relational databases as virtual RDF graphs. Translating RDF backwards to the relational model is developed in [12] under some assumptions. Conversion between

spreadsheet formats and RDF is also possible [6,10]. Google Refine is a power tool for working with messy data and generic format translations, with plugins supporting RDF.

Concept drift in the Semantic Web has been studied in [14], where the authors establish a theory for concept drift defining the meaning of a concept in terms of its intension, extension and labeling. Other Semantic Web approaches have used conceptual clustering [2] or concept signatures [5] to detect concept drift. In Description Logics, ontology diff [3] can be used to determine meaning differences. The question has been discussed in Philosophy around the confrontation of history of unit-ideas versus a pure linguistic intellectual history [9].

Some work has been done recently with respect to contexts in the Semantic Web, although they emphasize the specific goals of improving data integration [4] or speeding up reasoning [11]. Rule interchange languages for the Semantic Web like RIF are also related to dynamic concept construction [8].

2.3 Proposed approach

Format round-tripping. Existing approaches on format conversion pair any data format with RDF and perform a forward or backward transformation between the two. Our proposal is to take an holistic approach, studying the expressivity of these languages and checking whether arbitrary translation workflows are possible. We are interested in round-tripping translation paths to check if original representations can be regenerated without data loss. We aim at canonical RDF graph forms [1] and centric RDF data representations.

Concept drift. We will study what precise relationship holds between two different versions of a changing concept, identifying the presence of a drift and its nature. Using Description Logics work on ontology diff [3], we will define a minimum meaning concept core, which keeps stable over time despite other non essential transformations. A data model to represent drifted concepts will be needed. A systematic comparison between unstable concept properties will tell whether a drift occurred, and its type. We consider discussions on history of unit-ideas [9] and theories of concept drift for the Semantic Web [14] as inspiration.

Contextual knowledge. We will study how concepts can be dynamically defined depending on their graph contexts. To solve contextual knowledge questions we aim at a two step process. First, we target an explicit semantic representation of the context of a concept, and we will use various data models and vocabularies to define contexts. Second, we consider inference for deriving logical consequences from previously selected contextual graphs. This process can be further integrated with our concept drift framework.

2.4 Research methodology

We establish an iterative workflow that runs the proposed topics in parallel, first developing theories and then proposals. Proposals will be evaluated with at least the two Humanities case-studies referred in Section 1. All models and automated methods will

be validated by domain experts. At the end of each iteration, resulting design methods will be scaled up and refined.

2.5 Results and future work

Regarding format round-tripping, we implemented some preliminary tools^{1,2}. TabLinker is a MS Excel to RDF converter supporting translation of annotations and interactive user defined mappings. We also developed scripts generating RDF from various semistructured data formats. We plan to evaluate round-tripping by comparing an original file with its circularly translated homologue. With respect to concept drift, a first set of mappings between possibly label-drifted concepts have been defined using label similarity functions. We run simple longitudinal queries with MP2Demo, relying on an hybrid top-down/bottom-up approach that combines upper ontologies (e.g. Historical International Standard Classification of Occupations, HISCO) with automatically extracted local ontologies.

Three more yearly iterations will be carried out. Format round-tripping will be generalized from current scripts, defining transformation entities that will abstract specific format dependencies to modelling artifacts. We will create a data model for concept drift and an RDF/OWL simulation framework to test it with ontology diff and intension, extension and labeling functions. We will extend this framework to integrate reasoning with contexts.

2.6 Conclusion

In this research proposal we motivate the problems of format round-tripping, concept drift, and contextual knowledge in the context of a Humanities enabled Semantic Web. We propose an approach with novel perspectives extending the state of the art, and we describe an iterative research method to sort these issues

¹ <http://github.com/Data2Semantics/> ² <http://github.com/CEDAR-project/> out. Finally, we show work that has been done during the first year iteration, and we establish a plan for the remainder.

Chapter 3

Semantic Technologies in Historical Research

3.1 Introduction

Historians have a long tradition in using computers for their research [79]. The field of historical research is currently undergoing major changes in its methodology, largely due to the advent and availability of high-quality digital data sources. More recently, the Web has shaken the paradigm of research data publication, particularly since the inception of the Semantic Web [76] and the Linked Data principles [105]. This paper looks forward on how Semantic Web technology has been applied to historical data, and how these technologies can facilitate, boost and improve research by historians. This survey revisits the open problems in historical data and historical research, and analyses current contributions, namely papers, projects, online resources and tools, that apply semantic technologies to solve such problems. We study how successful these solutions have been and propose some challenges for the future.

Historical research is an interesting domain for the Semantic Web. Historical data are extremely context dependent, and always open to a variety of possible interpretations. Availability on the Web of historical research data, which concerns the study and understanding of our past, is growing. The Semantic Web is an evolution of the existing Web (based on the paradigm of the document) into a Semantic Web (based on the paradigm of structured data and meaning). It is not a separate Web but an extension of the current one, in which information is given well-defined meaning, better enabling computers and people to work in cooperation. This survey studies the crossroads of the Semantic Web and history as research domains.

We consider surveying the state of the art in Semantic Web and history a fundamental task for both fields. First, it is necessary as a knowledge organisation task, in order to articulate research and discern contributions. Second, it fosters development of semantic technology and history, both individually and as a unique field, and helps on building research agendas. Other attempts on gathering research efforts on Semantic Web and history exist, but most of them study specific history subfields [43, 144, 10]

or analyse concrete task-oriented tools [96, 14] and methodologies [105, 109, 110]. Moreover, none of them consist in surveys or literature reviews. To the best of our knowledge, this is the first survey reviewing contributions on history and the Semantic Web as generic fields of research.

The elaboration of the study in this paper is not free of obstacles. The first of them is the large amount of research contributions to survey, which had to be filtered to fit strictly the Semantic Web goals and the historical research goals in order to be feasible. By historical research we mean strictly research performed by historians, and talk about history as a research domain. Thus, we exclude other fields of the humanities in which historical research is also performed, such as art history or history of literature. Nevertheless, in the end the number of contributions amount to more than a hundred. Secondly, and even though the corpus of available literature is large, we also encountered difficulties on accessing some of the sources. To solve this, we combined the contributions with the knowledge of domain experts, conducting eight interviews with pioneers in this area. Third, structuring and articulating all this work is an arduous task that requires a lot of schemas, tables and discussions. Finally, the clash of the vocabularies used by two different research communities, usually pointing at similar issues, is problematic. To bridge different jargon we devote some space to cover existing classifications of historical data, especially discussing terms like *structure*, and we map historical data problems in terms of Semantic Web solutions.

The paper delivers four contributions. First, it describes a classification of historical data depending on several factors, merging existing distinctions by historians with structural approaches from computer science. Second, it articulates the research conducted in the emerging field of historical Semantic Web in terms of several *tasks*, and depicts the current landscape on advances in representing historical data with semantic technology. Third, we map the open problems of historical data with the solutions provided by the surveyed research. Finally, we show some open challenges for the future, considering first the not (or only partially) solved problems, and secondly Semantic Web facilities still to be explored in historical research.

While we concentrate on historical research, similar solutions emerge also in other humanities fields at the turn to e-humanities or Digital Humanities [78, 143]. As historical research overlaps with literary studies, ancient language studies, archaeology, art history and other humanities fields, these areas of encounter are also predestined candidates for the travel of generic methods developed from a semantic technology perspective for historical research to other humanities fields [123].

The survey is organized as follows. In Section 9.3 we introduce some background on historical research and the Semantic Web. In Section 3.3 we study the ecosystem of historical data. We describe the life cycle of historical information, propose a classification for historical sources, and show open problems of historical data. In Section 3.4 we articulate contributions that apply semantic technologies to historical research. In Section 3.5 we answer the question on how the contributions presented in Section 3.4 solve the open problems we describe in Section 3.3. In Section 3.6 we show the challenges that are still left to solve. Finally, in Section 10.7 we discuss our findings and establish some conclusions.

3.2 Background

3.2.1 Historical research

The field of historical research concerns the study and the understanding of the past. The field is currently undergoing major changes in its methodology, largely due to the advent of computers and the Web [79].

Computer science has inspired historians from the start. *History and computing* or *Humanities computing* were labels used before the inception of the Web [122]. Many pioneers in computer aided historical analysis have a background both in history and in informatics, and reflected early on about the usefulness of computational and digital techniques for historical research [79]. Ever since the advent of computing, historians have been using it in their research or teachings in one way or the other. The first revolution in the 1960s allowed researchers to harness the potential of computational techniques in order to analyze more data than had ever been possible before, enabling verification and comparisons of their research data but also giving more precision to their findings [66]. However this was a marginal group within the historical research: in general, the usage of computers by humanists could be described as occasional [93]. The emphasis was more on providing historians with the tools to do what they have always done, but now in a more effective and efficient way. Concretely,

- *databases and document management systems* facilitated the transition from historical documents to historical knowledge through text analysis;
- *statistical methods* were used predominantly for testing hypotheses, although with time were more valued as a descriptive or exploratory tool than as an inductive method;
- *image management* aided historians to digitize, enrich, retrieve images and visualize data [79].

Although computing tools are currently embedded in the daily life of most researchers, the use of these tools did not revolutionized all sciences equally. Accordingly, history failed to acknowledge many of the tools computing had come up with [79]. Instead of improving the quality of the work of historians and assisting them in their processes, software developed for historians often requires attending several summer schools [80]. Currently there are still many challenges and information problems in historical research. These difficulties mainly range from textual, linkage, structuring, interpretation, to visualization problems [79].

Despite these challenges, computing in history and in the broader sense the humanities, also brought some significant contributions in certain fields like linguistics (corpus annotations, text mining, historical thesauri etc..), archaeology (impossible without geographic information systems (GIS) nowadays), and other fields using sources that have been digitized for historical (comparative) research and converted to databases [79]. The use of electronic tools and media is incredibly valuable and important for opening up various sources for research which would otherwise remain unused. Open access to research data has always been an issue, especially in the humanities. However, over the past years various efforts have been made in opening up these black boxes

and making them available for researchers. These different sources contain rich information from various fields, which are often digital in nature in the form of databases, text corpora or images. These sources, in practice isolated databases, often contain a lot of semantics, but their data models were asynchronously designed, making them difficult to compare. So, while more and more sources are being digitized, more attention has to be given to the development of computational methods to process and analyze all these different types of information [104].

A key issue for historians and other humanities researchers when dealing with historical data for comparative research concerns the lack of consistency and comparability across time and space, due to changing meanings, various interpretations of the same historical situations or processes, changing classifications, etc.

Though not all research dreams materialized in the way initially envisioned [117], the inception of the Web allowed historians to aim for world-wide, large scale collaborations, especially in the area of economic and social history. This kind of web based cooperation allows to collect, distribute, annotate and analyze historical information all around the globe [89].

Changes in historical research are closely connected to the emergence of new scientific methods, and this co-evolution holds for decades and centuries. Statistics has influenced many fields including history, and paved the ground for quantitative studies [118]. However, these kind of historical studies became more and more the domain of sociologists, economists and demographers than scientists educated as historians [142]. Late important changes are consequences of recent technological trends connected to the emergence of the Web [132] and the inception of Semantic Web technologies [68].

3.2.2 The Semantic Web

The advent of the Semantic Web poses new perspectives, challenges and research opportunities for historical research. Envisioned in 2001 by Berners-Lee, Hendler and Lassila [76], the Semantic Web was conceived as an evolution of the existing Web (based on the paradigm of the document) into a Semantic Web (based on the paradigm of structured data and meaning). By that time, most of the contents of the Web were designed for humans to read, but not for computer programs to process meaningfully. Although computer programs could parse the source code of Web pages to extract layout information and text, computers had no mechanism to process the semantics. In other words, the Semantic Web *is not a separate Web but an extension of the current one, in which information is given well-defined meaning, better enabling computers and people to work in cooperation* [76].

More practically, the Semantic Web can be defined as the collaborative movement and the set of standards that pursue the realization of this vision. The World Wide Web Consortium [77] (W3C) is the leading international standards body, and the Resource Description Framework [162] (RDF) is the basic layer in which the Semantic Web is built on. RDF is a set of W3C specifications designed as a metadata data model. It is used as a conceptual description method: entities of the world are represented with nodes (e.g. *Dante Alighieri* or *The Divine Comedy*), while the relationships between these nodes are represented through edges that connect them (e.g. *Dante Alighieri wrote The Divine Comedy*). These statements about nodes and edges are expressed

as *triples*. A triple consists of a subject, a predicate, and an object, and describes a fact in a very similar way as natural language sentences do (e.g. subject: *Dante Alighieri*; predicate: *wrote*; object: *The Divine Comedy*). Subjects and predicates must be URIs (Uniform Resource Identifiers, the strings of characters used to identify and name a web resource like a web page), while objects can be either URIs or literals (like integer numbers or strings) [105]. RDF can be considered a knowledge representation paradigm where facts and the vocabularies used to describe them have the form of a graph. This setting makes RDF very suitable for data publishing and querying on the Web, especially when (a) the dataset does not follow a static schema; and (b) there is an interest of linking the dataset to other datasets.

Efforts on standardization have produced ontologies and vocabularies to describe multiple domains. An ontology is an *explicit specification of a conceptualization* [100] and contains the classes, properties and individuals that characterize a given domain, such as history. In the Semantic Web, the design of ontologies is done using the Web Ontology Language [159] (OWL). OWL consists of several language variants built upon different modalities of Description Logics [71] (DL), a family of formal knowledge representation languages. Such languages allow reasoning, that is, to extract or deduce consequences and new knowledge from the original.

A large number of RDF datasets have been published and interlinked on the Web, using these ontologies and vocabularies and following the Linked Data principles [75]. In the middle of the document-Web and the data-Web, formats and vocabularies for rich structured document markup (such as RDFa [161] or schema.org [39]) are enabling software agents to crawl semantics from web pages, bridging the gap between the Web for humans and the Web for machines. These efforts have evolved the Web into a global data space [105] where data can be queried e.g. using the SPARQL query language (SPARQL Protocol and RDF Query Language) [163]. Although the transition from the document-Web to the database-Web exists in the form of these standards and technologies, the simple idea of the Semantic Web remains largely unrealized [145].

3.3 Historical data

Since the introduction of computers in the field, historical research has produced high-quality digital resources [79]. Historical datasets encompass texts, images, statistical tables and objects that contain information about events, people and processes throughout history. Converted or born-digital, historical datasets are now analyzed at big scale and published on the Web. Their temporal perspective makes them valuable resources and interesting objects of study.

In this section we describe the ecosystem where historical information lives. First we introduce the life cycle of historical information, which is the framework we use to study how historical data is created, enriched, edited, retrieved, analysed, and presented. Then we propose a classification of historical data depending on several factors. Finally, we revisit the traditional open problems of historical data. Some of these problems have found solutions in current Semantic Web developments we present in Section 3.4.

3.3.1 The life cycle

The main object of study in historical research is historical information, and the multiple ways to create, design, enrich, edit, retrieve, analyse and present historical information with help of information technology. It is important to distinguish historical information from raw data in historical sources. These data are selected, edited, described, reorganized and published in some form, before they become part of the historian's body of scientific knowledge. We use the life cycle of historical information proposed by Boonstra et al. [79] to study the workflow of historical information in historical research.

Historical objects go through distinct phases in historical research. In each phase, these objects are transformed in order to produce an outcome meeting specific historical requirements. The phases can be laid out as the workflow of a *historical information life cycle* (see Figure 3.1). The phases, although sequentially presented, do not always have to be passed through in rigorous order; some can be skipped if necessary. The phases are also quite comparable with the practice in other fields of science.

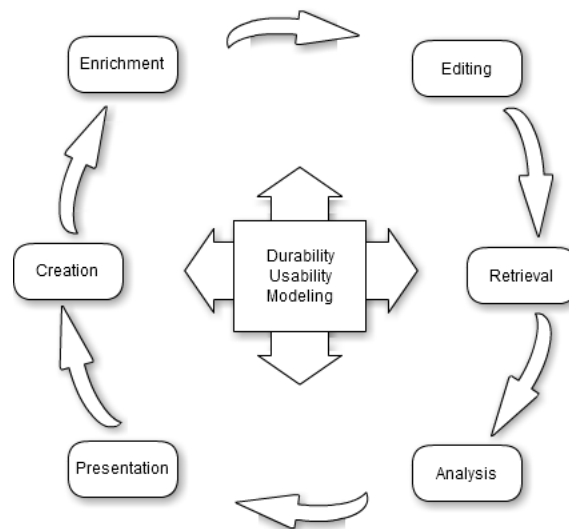


Figure 3.1: The life cycle of historical information (Boonstra et al. [79]). The phases in the life cycle are: (1) creation; (2) enrichment; (3) editing; (4) retrieval; (5) analysis; and (6) presentation.

The life cycle of historical information consists of six phases:

1. **Creation.** The first stage of the life cycle is the creation stage. The main aspect of this stage consists of the physical creation of digital data, including the design of the information structure and the research project. Examples of activities in this phase would be the data entry plan, digitisation of documents (through e.g. OCR), or considering the appropriate database software.
2. **Enrichment.** The main goal of this phase is to enrich the data created in the previous step with metadata, describing the historical information in more detail, preferably using standards such as Dublin Core [88], and intelligible to retrieval

software. This phase also comprises the linkage of individual data that belongs together in the historical reality, because these data belong to the same person, place or event.

3. **Editing.** Editing includes the actual encoding of textual information, like inserting mark-up tags or entering data in the fields of database records, with the intention of changing or adding historical data of convenience. All data transformations through algorithmic processes prior to analysis also belong to this phase. Editing also extends to annotating original data with background information, bibliographical references and links to related passages.
4. **Retrieval.** In this phase information is retrieved, that is, selected, looked up, and used. The retrieval stage mainly involves selection mechanism look-ups such as SQL-queries for traditional databases or Xpath [164] and Xquery [165] for XML-encoded texts.
5. **Analysis.** Analyzing information means quite different things in historical research. It varies from qualitative comparison and assessment of query results, to advanced statistical analysis of data sets.
6. **Presentation.** Historical information is to be communicated in different circumstances through multiple forms of presentation. It may take very different shapes, varying from electronic text editions, online databases, virtual exhibitions to small-scale visualizations. It can happen frequently in other phases as well.

In the middle of the historical information life cycle, three aspects are identified which are central to history and computing, but also in the humanities in general:

- *Durability* ensures the long term deployment of the produced historical information.
- *Usability* refers to the ease of efficiency, effectiveness and user satisfaction.
- *Modeling* denotes to more general modeling of research processes and historical information systems.

3.3.2 A classification of historical data

The continuous usage of computing in different areas of historical research has produced digital historical data with different formats, perspectives and goals. To be used in the Semantic Web, these historical data have to be represented semantically, using the current standards (see Section 3.2.2). In this section we propose a classification of historical data in order to bridge the gap between the data representation tradition in historical research, and the standard modelling paradigms of the Semantic Web [68, 105].

Primary and secondary sources

Historical sources can be characterized and divided in many ways. A basic distinction used by historians is between *primary* and *secondary* sources.

Primary sources are original materials created at the time under study [74]. They present information in its original form, neither interpreted, condensed nor evaluated by other writers, and describe original thinking and data [70]. Examples of primary sources are scientific journal articles reporting experimental research results, persons with direct knowledge of a situation, government documents, legal documents (e.g. the Constitution of Canada), original manuscripts, diaries (e.g. the Diary of Anne Frank) and creative work. Primary sources can be distinguished into *administrative* sources and *narrative* sources, like biographies or chronicles. Administrative sources contain records of some administration (census, birth, marriage and death rolls, administrative accounts of taxes and expenses, resolutions minutes of administrative bodies, deeds, contracts, etc.). Typically, historians want to extract the facts in order to gather statistical data. Narrative sources are full text documents containing a description of the past, made by an author being an eyewitness: think of diaries, chronicles, newspaper articles, diplomatic reports, political pamphlets, etc. Historians may be interested in both, factual information and the author's vision and the bias.

Secondary sources are materials that have been written by historians or their predecessors about the past [153]. They describe, interpret, analyze and evaluate the primary sources. Usually, secondary sources gather modified, selected, or rearranged information of primary sources for a specific purpose or audience [70]. Examples of secondary sources are bibliographies, encyclopedias, review articles and literature reviews, or works of criticism and interpretation.

Since historical data have not been produced under the controlled conditions of an experiment, historical research always has something of the work of a detective, and certain details (read: annoying inconsistencies) cannot be destroyed or manipulated. These details may contain relevant information. On the other hand, to be able to extract statistical information and come up with more general statements, some formalization, relating information and harmonizing expressions of what is later used as variables is needed. Harmonization, the process of making data-sources uniformly accessible without altering its original form, is closely related to issues of standardization and formalization [125].

Intended further processing

Some historians [79] propose to structure historical data depending on their required further machine processing. They distinguish between *textual data*, *quantitative data* and *visual data*. Textual data comprises the whole set of unstructured historical sources, such as letters, memoranda or biographies, all in a form of free text. Quantitative data can be seen as historical sources aiming at a quantitative analysis, like church registers, census tables and municipality micro-data. Finally, visual data gathers all kinds of historical evidence not encoded by text or numbers, such as photographs, video footage and sound records.

Source oriented vs. goal oriented

Researchers make the distinction between *source oriented* and *goal oriented* historical data [79]. When dealing with historical data it is important to decide in an early stage whether the data should be modeled according to a source or goal oriented approach. The source oriented approach aims to postpone enforcing any standards or classifications, resemble the underlying source data as close as possible (schema free representation) and hence allow room for multiple interpretations of the data. Another approach is the goal or model oriented approach. Historical data is often plagued with inconsistencies, changing structures and classifications, redundant or erroneous data and so forth. The goal oriented perspective therefore advocates the use of more sound data models to start with. This means restructuring the data according to certain views or goals which are mainly dependent on expert knowledge. Accordingly, this perspective commits to a certain data model in an early stage.

Level of structure

At the end of the creation phase (see Section 3.3.1) one may expect to have a historical dataset suitable for further processes. However, the nature of the steps to be taken thereafter may strongly depend on the way the resulting dataset is structured. Indeed, attaching Semantic Web technologies to these historical sources (e.g. to extract RDF triples from them, or enrich them semantically) is strongly dependent on their level of structure. We propose the historical data classification shown in Figure 3.2. We distinguish three levels of inner structure in historical datasets: *structured*, *semi-structured* and *unstructured*. Each level of structure can be divided into several *types of structure*.

Structured data. Structured data refers to sources that have a clearly defined data model. A data model is an abstract model that documents and organizes data for communication, and is used as a plan for developing applications. An example of a structured dataset would be census material published in rows and columns, or a database of historical events. Well known generic examples of such a structure are sources encoded as relational databases, XML files, spreadsheet workbooks or RDF datasets. It is easy to see that all these examples meet a certain abstract model for the data they represent (relational schemas, DTD constraints, tabular formats and RDF triple statements). Structured historical data are usually managed with *relational databases*, *graph/tree representations* and *tabular representations*. Relational databases are the most well-known way of committing to some schema for representing historical objects and their relationships. Because their structure, relational databases are ideal for goal or model-oriented representation of historical data [79] with some concrete conception of reality in mind.

Relational databases. Relational databases have their own languages (SQL) and systems (MySQL, Microsoft SQL Server, PostgreSQL, Microsoft Access, Oracle, etc.) to represent and store historical data. They all follow the relational model [85]. Some issues, especially when trying to integrate data from different databases and modelled with different conceptual schemas, appear often in historical datasets encoded this way.

Graph/tree representations. Relying on graph theory, graph databases offer mechanisms for storage and retrieval of data with less constrained consistency models than

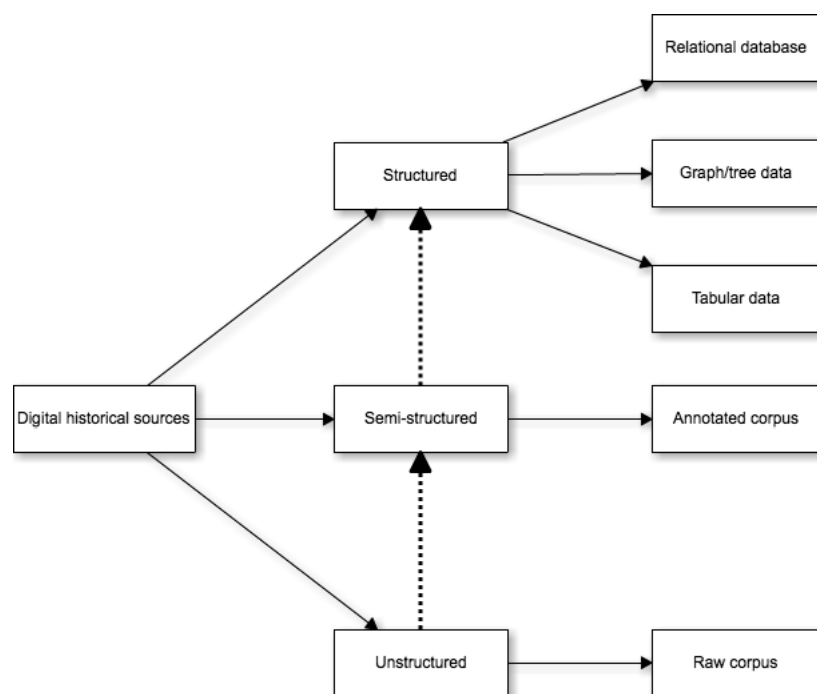


Figure 3.2: Classification of historical data according to their level of structure. Dotted arrows indicate the direction of usual transformations in workflows that identify historical entities (and their relations), from unstructured to structured representations.

traditional relational databases. They provide variable performance and scalability, but high flexibility and complexity support. AllegroGraph, IBM DB2, OpenLink Virtuoso and OWLIM are typical examples. To exchange historical data in graph form, RDF (see Section 3.2.2) is used. Graph/tree data is found in historical samples that come in formats such as XML (trees), RDF (graphs) or JSON (JavaScript Object Notation). Although they are conceived for modeling data in very disparate models (a tree, a graph and nested dictionaries, respectively) and purposes (e.g. JSON is mainly used for data interchange between web applications and services), these formats also follow some assumptions to put structure on historical data.

Tabular representations. Some historical datasets are encoded in tabular form. Tables consist of an ordered set of rows and columns, the latter typically identified with a name. The intersection of a row and a column is a cell. Depending on the specific encoding (Comma-separated values (CSV), Microsoft Excel spreadsheets, etc.) tables can offer variable features. Tables are used to store all kinds of historical data, especially meso, macro and microdata about individuals, registries, or historical population censuses.

Semi-structured data. Semi-structured data appear more often as an intermediate representation between unstructured and structured historical data than as raw historical data. Typical technologies applied here are markup languages, such as XML, to denote special characteristics of historical texts in specific regions of the corpus. *Annotated corpora* are the most important example of semi-structured data. They usually consist of raw historical texts with annotations on well-defined text regions, usually implemented with a markup language, like XML.

Unstructured data. In case a data model such as we described for structured data does not exist, we talk about unstructured data. In unstructured data there is scarce or no structure at all. The typical example is unconstrained, raw corpora encoded in plain text files. Unstructured sources are the most common representation of historical data, typically transcriptions of historical texts. Objects with a high variety of historical nature can be included in this category: letters, books, memoranda, acts, etc.

The use of the terms *structured* and *unstructured* in computer science to describe datasets is very different from the use of those notions in history, where administrative sources are often labeled as *structured* and the textual secondary sources as *unstructured*. Also narrative sources have internal structures, which can be made explicit. From the 19th century onwards historians have made scholarly source editions, which contain structured and annotated information. Nowadays the printed source editions are replaced and supplemented by databases and XML-based digital editions. So, *structured* or *unstructured* are relative notions: administrative sources usually have an obvious structured layout, while narrative sources have a latent, at first sight *hidden* structure, which is made explicit as soon as they appear in a scholarly source edition. So, both administrative and narrative sources can appear in the form of *structured* or *unstructured* data in computer science jargon.

Discussion

Although structure really matters for deciding what specific computing technique or semantic model has to be applied to the sources, being those sources administrative

or narrative, deliberate or inadvertent, does not really matter if their inner structure is clearly identified. Their belonging to one type or another may have an influence at some point, but in general the procedure to extract RDF triples from the sources strongly relies on the type of source we have regarding their structure. The goal is a faithful representation of the source in Semantic Web formats: a source-close representation allowing to model data as-is, meeting the same requirements of faithfulness than critical source editions (which is the standard for historians). It is critical for semantic representations to consider *context* and *source structure* as critical editions do, because they may be relevant for interpretation of the data. A digitized, semantically-enabled historical source should ideally preserve context and structure and support goal-oriented extraction of data, in order to construct historical facts in the framework of a certain research. By means of dataset interlinking and appropriate design and usage of ontologies and vocabularies, *context* and *source structure* should be able to be preserved using semantic technologies. To this end, ontologies can be contextualized to conciliate a party's subjective view of a domain [81].

3.3.3 Open problems

The classification proposed in Section 3.3.2 is not strict and admits hybrid examples. For instance, annotated digital text sources can be provided both as XML files or stored in a relational database (e.g. for statistical analysis). Some authors classify sources that combine primary and secondary sources like these as tertiary sources [134].

Although many advances have been made in different fields and computers are seen as valuable assets, a high percentage of historians are unfamiliar with or remain unconvinced that semantic technologies may become a new methodological asset [66, 149]. The reason is that the weapon of choice of historians was and remains mostly the database, particularly in relational form [66]. This not only enabled historians to retain some of the integrity of the original data sources, but also paved way for rapid advances on issues such as classifications and record linkage. Therefore, historians typically do research using their *own* datasets, resulting in the creation of a vast amount of scattered data and specific technological challenges. In this section we revisit the traditional open problems of historical research derived from this tradition. In Section 3.5, after presenting the state of the art in the application of semantic technologies to historical research in Section 3.4, we point to the specific open problems described here that are eventually solved with such technologies.

Historical data problems can be divided into four main categories: information problems of historical sources, information problems of relationship between sources, information problems in historical analysis, and information problems of the presentation of sources [79].

Historical sources

The first set of open problems in historical research happens in phase 1 of the historical data life cycle (see Section 3.3.1). This is when the historical data are created.

Manually encoded or OCR-scanned, the creation of the dataset reveals the first barriers. Some characters, words or entire phrases in the original material may be lost

or impossible to read or recognise by the human or the computer. Moreover, different techniques may extract historical entities differently. An example would be: what is the word that is written on this thirteenth-century manuscript?

The next question usually is: what does it mean? Background knowledge is provided by libraries in the offline world. But the computer aiding tools also need to have means to help the historian, using the Web as channel and semantics as meaning.

Related to background knowledge is the provenance of the data. Even if the source is clearly identified and its meaning deciphered, the historian needs to know more. To which issue does it relate? Why was it put there? Why was the text written? Who was the author? Who was supposed to read the manuscript? Why has it survived?

Another main issue relates to the structuring problem of historical data [138]. How can historical objects be encoded in a database? Researchers have to decide on what is an adequate data model for their datasets. As historians often have no clear research question when starting an investigation, it is neither possible nor desirable to model the data according to certain requirements in advance. Moreover, different sources have been produced throughout different periods in history with different views and motives. Historical census data is a good example, having varying structures and changing levels of detail which hinders comparative social history research both in past and present efforts [138].

The main discussion regarding this involves whether to use a source or a goal oriented data model for historical data (see Section 3.3.2). Researchers in favor of the source oriented approach claim that a commitment to a certain data model suitable for analysis should be postponed to the final stages of a project, in order to maintain flexibility and build on the data in a non destructive manner. This is especially the case when the database is supposed to be shared with other researchers or used in the future [121].

Relationships between sources

As historical researchers deal with various isolated sources, they face the problem of how to integrate these dissimilar sources for their purposes. This typically happens in phase 2 (enrichment) of the life cycle of historical information (see Section 3.3.1). An example would be: is this Lars Erikson, from this register, the same man as the Lars Eriksson from this other register?

Quite often several sources are used in historical research, which makes linking different sources another key problem. Micro data of the same person contained in different censuses, parish registers, marriage or death certificates are a good example. Obvious linkage problems are how to disambiguate between persons with the same name, how to manage changing names (e.g. in case of marriage of a woman) and how to standardize spelling variations in the names. In databases, several issues affect data comparability. *Schema mismatch* occurs when two different databases cannot be compared because of semantic differences in the concepts of their defining schemas. For instance, two XML files conformant to different DTD schemas may define and structure differently the same historical entity. Additionally, *value mismatch* occurs when the allowed values for columns or variables in two databases are different. It may also happen across datasets despite being schema or vocabulary-compatible. For

instance, an attribute may encode the variable *social class* with categories *A, B, C*, while other dataset may do so with categories *high, medium, low*.

Other problems relate to how to link historical data with their spatial and temporal context. For example, some historical facts may need to be linked with occupational titles that evolve over time [22] or with countries with changing geographical boundaries (compare for example the contemporary geographic position of countries in Europe with the situation in 1930 and in 1900; or the fact that the city of Rotterdam suffered nine major changes in its composition between 1886 and 1941 [65]). As historical research often deals with changes in time and space, historians require tools which enable them to deal with these aspects. Accordingly several techniques have been developed for historical research, but the applicability of these has yet to be determined [79].

Historical analysis

Historical analysis is a fundamental part of the life cycle (see phase 5 in Section 3.3.1). It usually implies data transformations that aid historians in guiding their research. It also builds the bridge between their hypotheses and historical evidence.

The first issue in analysis is the massive treatment of historical data processed in previous stages to satisfy historical requirements, or to support a specific historical interpretation. An example would be: from this huge amount of digital records, is it possible to discern patterns that add to our knowledge of history? Various statistical techniques are borrowed from the social sciences to this end, like multilevel regression, and other techniques have been specifically developed for historical research, such as *event history analysis*. However, addressing historical data analysis in a broad sense remains essentially unsolved.

In historical research the meaning of data cannot exist without interpretations [79]. Due to drifting concepts in history, different interpretations could exist with regards to certain data [167]. However as interpretation of data is a subjective matter, this information should be added in a non destructive way, preserving the original source data.

Presentation

Presentation is the final phase of the historical information life cycle (see Section 3.3.1). Its goal is to use visualizations to aid the study and comprehension of historical data. An example problem of such phase would be: how do you put time-varying historical information on a historical map?

Presentation of historical data must be adequate. Different types of presentations are suitable at different stages of a research project. Presentation may take different shapes, varying from digitized documents, poorly and well modelled databases, or visualizations and representations on Geographic Information Systems (GIS). Currently there is a great need for tools and methods to present changes over time and space.

3.4 Findings

In this section we review the current state of the art in the application of semantic technologies to historical research, describing relevant contributions towards a historical Semantic Web.

The contributions are classified in the categories of *scientific papers*, *research projects*, *online resources* (presentations, online articles), and *tools, ontologies and lexical resources* (ontologies, demos, applications or programming libraries). Tables 3.4, 3.1, 3.2 and 3.3 show these contributions. Additionally, we map each contribution to one or more specific *tasks*. These tasks are shared areas of concern for both historical research and the Semantic Web. We identify the following four tasks: *knowledge modelling*, *text processing and mining*, *search and retrieval*, and *semantic interoperability*. We develop the work on each task in the following sections. We group contributions that cover the full spectrum of tasks under a final section in *longitudinal approaches*.

3.4.1 Knowledge modelling

Under this category we study research that has been conducted to model historical knowledge or historical facts using standard Semantic Web representations (see Section 3.2.2). We group contributions to a semantically enabled historical web by the following emphasis of research: historical ontologies, and linking historical data.

Historical ontologies

Data models are necessary for giving structure to any historical data, since they are the abstract models that document and organise data properly for communication. Ontologies encode such models in the Semantic Web [76] (see Section 3.2.2), and attention has been given to the need of historical ontologies [110]. In historical research, ontologies are the providers of metadata and background knowledge in phases 2 (enrichment) and 3 (editing) of the historical information life cycle (see Section 3.3.1). Semantic Wikis [139, 42] are a great resource for historians to collaboratively build such ontologies.

We find a first category of such models in the form of (typically XML-encoded) taxonomies for historical research. A taxonomy is a collection of controlled vocabulary terms organized into a hierarchical structure, in general with less expressivity than an ontology. The first important example of such knowledge organization is the CLIO system, a databank oriented system for historians [151] appeared in 1980. CLIO included a tag/content representation for historical data that could be structured in complex hierarchies, supporting the recoding of material with doubtful semantics. CLIO remained as *the* system for organizing historical knowledge until the inception of the Web.

More recently, the Semantic Web for Family History [57] exposes a set of genealogy markup languages based on XML to semantically tag genealogical information on sources containing that kind of historical data. In the context of the Text Encoding Initiative [52] (TEI) there is an important discussion about building the bridge between

Project name	Knowledge modelling	Text processing & mining	Search & Retrieval	Semantic interoperability
Agora [1]	○	✓	✓	
BRIDGE [5]	✓		✓	○
CHoral - access to oral history [8]	✓		✓	○
Historical Timeline Mining and Extraction (HiTiME) [24]	○	✓	✓	○
LINKing System for historical family reconstruction (LINKS) [27]	✓			✓
SCRipt Analysis Tools for the Cultural Heritage (SCRATCH) [40]	○	✓	✓	
FDR Pearl Harbor Project [15]	✓	✓	✓	✓
North Atlantic Population Project (NAPP) [32]	○		○	✓
Circulation of Knowledge and Learned Practices in the 17th-century Dutch Republic (CKCC) [9]		✓	○	○
Voyage of the Slave Ship Sally [59]	○	✓	○	
Multilingual Access to Large Spoken Archives (NSF-ITR/MALACH) [33]	○	○	✓	✓
H-BOT [20]	○	✓	✓	
Clergy of the Church of England Database (CCed) [6]	✓		✓	○
Armadillo: Historical Data Mining [2]	✓	✓	✓	✓
Historical Event Markup and Linking (HEML) [21]	✓		✓	
SAILS [38]	✓		✓	✓
CLARIN-Verrijkt Koninkrijk [10]	✓	✓	✓	✓
Historical International Standard Classification of Occupations (HISCO) [22]	✓			✓
Historical Sample of the Netherlands (HSN) [23]	○		✓	✓
CEDAR [7]	✓	○	✓	✓
Linking History in Place [26]	○		✓	✓

Table 3.1: Reviewed projects. The ✓ and ○ signs indicate a strong and a medium relationship, respectively, between the contributions (rows) and the tasks (columns).

Resource name	Knowledge modelling	Text processing & mining	Search & Retrieval	Semantic interoperability
Semantic Web approaches in Digital History: an Introduction [43]	✓			✓
Fawcett: A Toolkit to Begin an Historical Semantic Web [14]	✓	✓	✓	✓
Spatial cyberinfrastructures, ontologies, and the humanities [46]	✓	✓	✓	✓
SIG Ontologies [44]	✓			✓
CultureSampo - Finnish Culture on the Semantic Web 2.0: Thematic Perspectives for the End-user [11]	✓	✓	✓	✓
Text Mining for Historical Documents: Topics and Papers [53]	○	✓	○	
RDF vocabularies for historic placenames and relations between them [37]	✓			✓
The Semantic Web for Family History [57]	✓		✓	✓
Data portal for Social Sciences. Open data with SPARQL endpoint [12]	✓		✓	✓

Table 3.2: Online resources. The ✓ and ○ signs indicate a strong and a medium relationship, respectively, between the contributions (rows) and the tasks (columns).

Tool, ontology, lexical resource	Knowledge modelling	Text processing & mining	Search & Retrieval	Semantic interoperability
NLP2RDF [31]	○	✓		
SIMILE/Timeline [45]	○		✓	
Gapminder [18]			✓	✓
TokenX [58]	○	✓		
TAPoR [51]	✓	✓		
SEM event model [41]	✓			○
OpenCYC [34]	✓			✓
XCES [63]		✓		✓
Dublin Core [88]	✓			○
GATE [19]		✓		
WordNet [62]	○	✓		
Framenet [16]	✓	✓		
SUMO [48]	✓			○
MILO [29]	✓			○
AskSam [3]		✓		
TEI (Text Encoding Initiative) [52]		✓		✓
SGML [47]		✓		✓
TACT [50]		✓	○	
Wordcruncher [61]		✓		
Atlas.ti [4]		✓	○	
NLTK [30]		✓		
FRED [17]	✓	✓		
WAHSP and BILAND [60]	○	✓		
The Event Ontology [55]	✓			○
LODE [28]	✓			○
Semantic MediaWiki (SMW) [42]	✓		✓	✓
Europeana Data Model (EDM) [13]	✓			○

Table 3.3: Tools, ontologies, and lexical resources. The ✓ and ○ signs indicate a strong and a medium relationship, respectively, between the contributions (rows) and the tasks (columns).

Paper title	Knowledge modelling	Text processing & mining	Search & Retrieval	Semantic interoperability
Hacking History via Event Extraction [144]	○	✓	✓	
Exploiting Semantic Web Technologies for Intelligent Access to Historical Documents [109]	○	✓	✓	
Historical Ontologies [110]	✓			○
Virtual Knowledge in Family History. Visionary Technologies, Dreams and Research Agendas [117]				✓
Past, present and future of historical information science [79]	○	○	○	✓
Proposed category system for 1960-2000 Census occupations [130]	○			✓
The Comparability of Occupations and the Generation of Income Scores [148]	○			✓
Challenges and Methods of International Census Harmonization [91]	○			✓
Making Sense of Census Responses: coding complex variables in the 1920 PUMS [98]	○			✓
Semantic Networks and Historical Knowledge Management: Introducing New Methods of Computer-based Research [112]	✓	○		
Queries in Context: Access to Digitized Historic Documents in a Collaboratory of the Humanities [152]		○	✓	
Converting a Historical Architecture Encyclopedia into a Semantic Knowledge Base [169]	○	✓	✓	
Historical documents as monuments and as sources [86]		○	✓	
Digital Hermeneutics: Agora and the Online Understanding of Cultural Heritage [156]	○	✓	✓	
Visualizing an Historical Semantic Web with	✓		✓	

XML (taxonomies) and OWL (ontologies) in historical data. SIG: Ontologies [44] contains a full log on contributions on how to use ontologies with TEI formats; namely, how TEI-XML encoded documents can refer to historical concepts and properties that have been previously formalized in an external OWL ontology.

The Historical Event Markup and Linking Project [21, 141] (HEML) was probably the first project with the goal of creating a Semantic Web of history. Started in 2001, it explored the use of W3C markup technologies to encode and visualize historical events on the Web. Although in the beginning XML was the selected language to provide tagging and markup for describing historical events, the project later experimented with RDF to model and visualize them [140]. This transition was also happening in the whole historical ontologies community, as researchers better understood RDF and its differences with XML.

The modelling and representation of events, often defined as *persons* doing an *activity* in a certain *place* and *time*, has received a lot of attention in the development of historical ontologies, and most practical results show that the concept of the event is at the core of historical knowledge modelling. Van Hage et al. [157] design the Simple Event Model [41] (SEM), intended to model events in the domains of history, cultural heritage, multimedia and geography. Similarly, the Event Ontology [55], inspired in the musical domain, models the representation of events as combinations of persons, places, moments in time, and factors. Finally, LODE: An ontology for Linking Open Descriptions of Events [28] is especially intended for the publication of historical events as Linked Data. Interestingly, these ontologies have a great overlap in their conceptual modelling of events even coming from different domains. On the other hand, some studies point out specific modelling needs for different historical domains, stressing that historical ontologies should reflect how a particular time frame influences the definitions of concepts [110].

Another big focus in historical ontologies is given to geographical modelling. Owens et al. [136] describe a geographically-integrated history, and stress the importance of dynamics and semantics in Geographic Information Systems (GIS). They set an agenda for historical GIS systems that includes important semantic modelling tasks involving ontologies and geography for historical analysis. Moot et al. [131] depict the interesting crossroad between text analysis, historical semantics and geography in a work that structures geographical knowledge from a historical corpus of itineraries. Vocabularies for historical place names are under discussion [37]. Although not intended for historical research, the GeoNames ontology [56] is the reference for geographical modelling in the Semantic Web.

Since entities like places, persons or events change their over history and time, there is work raising the importance of a change-aware modelling in ontologies [94, 123, 127]. In historical research and the Semantic Web this is especially true for geographical names, places and regions [108], but also for demographical, social and economical indicators, such as occupations [22].

Linking historical data

By understanding the use and advantages of semantic technologies, practitioners and researchers of historical data can not only connect their own data sources but more-

over, also disseminate their data into the Semantic Web and integrate it with other data sources which were previously not possible or cumbersome. The approaches reviewed in this section match the historical data problem of the *relationships between sources* (see Section 3.3.3). In most cases, the use of semantic technologies solves it.

If one side of knowledge modelling stresses the importance of ontologies and formalization of the semantics of historical domains, the other side pursues the usage of such ontologies to interlink related historical data on the Web. Some researchers in history have centered their interest in how semantics can help relating and linking historical sources and entities: *historical, semantic networks are a computer-based method for working with historical data. Objects (e.g., people, places, events) can be entered into a database and connected to each other relationally. Both qualitative and quantitative research could profit from such an approach* [112]. Linking historical datasets appropriately is an old and very well known problem in historical research [79]. The landscape on current projects linking historical data (typically extracted from unstructured sources) shows a tendency on publishing more and more historical Linked Data in RDF.

There is a wide variety of project types looking for that structure, though not doing so solely (or explicitly) in RDF. For instance, the Circulation of Knowledge and Learned Practices in the 17th-century Dutch Republic [9] (CKCC project) studies the epistolary network for circulation of knowledge in Europe in the 17th century, extracting all entities and links from the correspondence of scientific scholars of that time. The LINKing System for historical family reconstruction (LINKS) project [27] reconstructs the links between individuals of historical families across several registries. The CCed [6] project follows a similar approach with clerical careers from the Church of England Database. While these projects mine the historical sources for important historical personalities and their relationships, other approaches, such the SAILS [38] project, dive into more concrete historical events and links various World War I naval registries together. The common goal in these initiatives is to produce a *semantic network of historical data containing objects like people, places and events connected to each other*, which clearly matches the intended purpose of historical ontologies (see Section 3.4.1), but also the general mission of the Semantic Web [76] and Linked Data [105].

Many other projects expose their domain specific historical datasets using RDF. These datasets facilitate their linkage to others using existing ontologies (see Section 3.4.1), achieving shared goals with the old task of historical record linkage. For instance, the Agora project [1] aims at formally describing museum collections and linking their objects with historical context using the SEM [41] (Simple Event Model). Historical events are found elsewhere in historical data. The FDR Pearl Harbor project links events, persons, dates, and correspondence found on government letters and memoranda on the surroundings of the Pearl Harbor attack on 1941 between the US and Japanese governments. All these entities are represented in RDF to model a graph of historical knowledge about that particular event. From a more socio-historical point of view, the Verrijkt Koninkrijk [10] project links RDF concepts found on a structured version of De Jong's studies on *pillarization* of the Dutch society after the World War II. More focused on media, the Poli Media project [36] mines the minutes of the general state debates to link historical entities to the archives of historical newspapers,

radio bulletins and television programs. The goal is to create a unified historical search environment, facilitating a cross-media analysis [114].

Some general purpose tools facilitate the creation of historical Linked Data. The Fawcett toolkit [14] and the Armadillo project [2] are good examples. The latter exports RDF from any unstructured historical source, producing an RDF graph of historical knowledge that encodes the historical entities and their relationships expressed in that source. Other tools like Open Refine [35] or TabLinker [49] are tailored to produce such Linked Data from structured sources like tables (see Section 3.3.2).

3.4.2 Text processing and mining

In *text processing and mining* we revise work that deals with processing unstructured text. Textual resources play an important role in history research. We especially survey work on automatically extracting historical entities (such events or persons) via Natural Language Processing (NLP) techniques. The purpose of NLP is to enable computers to derive meaning from human, natural, or unstructured language input (see Section 3.3.2).

Structuring historical information from textual resources for further analysis is the bottom line of many research projects. The interesting differences come usually from the various source materials these projects mine. The general public-aimed Agora project [1] enriches museum collections with historical knowledge in order to help users place museum objects in their historical contexts. To this end, Agora employs information extraction techniques from statistical natural language processing to extract named entities (actors, locations, times, event names) from textual resources such as Wikipedia and collection catalogues which are used to populate SEM [41] (see Section 3.4.1) instances. From the object descriptions, also relevant historical entities are extracted which can be linked to the events. To formalize this workflow, Segers et al. [144] present a prototype extraction pipeline for extracting events and their properties from text using off-the-shelf natural language processing tools such as named entity recognition and pattern-based approaches. The main problem they encounter is that the notion of events is still ill-defined in NLP research, and as such tools are not yet readily available.

Textual encoding of the media have also been the source to extract historical knowledge in several projects. The Bridge project [5] aims at bringing more cohesion into Dutch television archives by finding relevant links between the official archives maintained at the Netherlands Institute for Sound and Vision and other information sources such as program guides and broadcasting organizations websites. It is thus focused on improving access to television archives for media professionals. In order to do so, relevant entities are extracted from archives by using statistical NLP techniques. Furthermore, they will detect interesting events in television archives by detecting redundant stories, utilising the structure of the archive to identify links between different entities [82]. The Poli Media project [36] mines the text of minutes of the general state debates to extract and link historical entities from the archives of historical newspapers, radio bulletins and television programs.

The Historical Timeline Mining and Extraction (HiTiME) project [24] is aimed at detecting and structuring biographical events. To this end they analyze biographies

of persons from the Dutch union history to create timelines that tell the life story of these persons, and social networks of the persons they interacted with. Van de Camp and Van den Bosch [155] describe an approach to build networks of historical persons by mining biographies for person names and relationships between persons. They use standard named entity recognition tools and utilise the inherent structure of biographies (the topic of the biography is a particular person, and any persons mentioned in this biography should have something to do with this person) to detect interpersonal relations.

Many ehumanities and ehistory projects are exploring document summary techniques or document enrichment techniques from NLP to aid search in their archives. One of these techniques is topic modelling, which can be used to add topic indicators to a document, which may help cluster search results or create more fine grained indexes of archive records. Wittek and Ravenek [170] explore the state of the art in topic modelling techniques to index 19,000 letters of correspondences between 16th and 17th century Dutch scientists.

Other high-level text analysis methods, such as frequency-based corpus analysis to compare e.g. work from different authors or investigation of other stylometry characteristics, are also popular in the ehumanities domain [54]. These methods are not domain-dependent and fit more easily into the ehumanities researcher search-based toolbox.

The spectrum of tools to extract knowledge from unstructured historical data is wide. Important contributions are essentially domain-independent [69], thus not particularly focused on historical text processing. Gangemi [96] presents a recent and complete comparison of generic knowledge extraction tools for the Semantic Web, which will aid historical researchers working in the phases 2 (enrichment) and 3 (editing) of the historical information life cycle (see Section 3.3.1).

3.4.3 Search and retrieval

In *search and retrieval* we include systems that exploit semantic formalisms as a new way of indexing, querying and accessing historical data, instead of relying on the traditional text-based or keyword-based algorithms. This task matches the phase 4 (retrieval) of the life cycle of historical information (see Section 3.3.1).

It is not a coincidence that a high number of contributions that aim at extraction of structured entities from historical data also point at some desired system able to improve search and retrieval of such entities. Indeed, by means of constructing a semantic graph of historical knowledge, search and retrieval of that knowledge, as well as indexing systems that give exact pointers to the source in which particular historical entities are mentioned, can be easily built and improved. The Agora [1] (museum collections), BRIDGE [5] (historical TV metadata), CHOoral [8] (historical audio metadata), Historical Timeline Mining and Extraction (HiTiME) [24] (biographical events), Verrijkt Koninkrijk [10] (Dutch post-war social clusters concepts) and FDR Pearl Harbor [15] (historical events around Pearl Harbor attack on 1941) projects are all good examples of this tendency. Once the knowledge is successfully extracted from the historical sources and formalized appropriately, entities structured this way can be used for a graph-based search and retrieval, for instance through SPARQL queries (see Section

3.2.2), although most systems use specific access methods [109]. Other projects, like the H-BOT [20] project, use a natural language interface instead of a query system for retrieval of such historical structured knowledge.

Indexing of historical contents is another way of improving search and retrieval of historical data. Indexing and historical data storage systems have a long tradition in historical research [79]. CLIO [151] is a traditional example of such a system, nowadays indexing is performed by XML annotation-oriented approaches, such as described by Robertson [141]. These initiatives should consider the emerging RDFa, microformats and microdata technologies (see Section 3.2.2) to study the ways they fit in the vast domain of historical text annotation systems.

3.4.4 Semantic interoperability

In this section we analyze to what extent contributions consider the problem of data integration and use the Semantic Web to deal with it. The specific problems encountered are data model mismatching, schema incompatibilities and disparate source formats. Semantic interoperability has much to do with data integration, namely, how to commonly query and uniformly represent data that come from multiple sources (i.e. fitting several, probably non-compatible data models).

Semantic heterogeneity of historical sources is especially present on social history projects. The North Atlantic population project [32] (publication of microdata of several Atlantic countries) has this problem of data harmonization, in which heterogeneity of sources requires an intense work on resolving data model inconsistency between datasets.

The source material for the Historical Sample of the Netherlands [23] (HSN) database consists mainly of the certificates of birth, marriage and death, and of the population registers. From those sources the life courses of about 78.000 people born in the Netherlands during the period 1812-1922 have been reconstructed. Stored in a database and downloadable as files, this information forms a unique tool for research in Dutch history and in the fields of sociology and demography. As in the case of the HSN this type of sources is usually stored in archives, and, for the majority from a more remote past, not yet machine readable and not easy to analyse with NLP techniques. There is one major pitfall in linking this kind of data: extracting data about persons, events, institutions, locations is one thing, but linking to their different instantiations (for instance different name spellings, or persons with the same name) and keeping good documentation is the real challenge [121].

The CEDAR project [7], located in the crossroads of the Semantic Web, statistical analysis and social history, exposes the Dutch historical census data in the Semantic Web. Censuses are a great source of non-biased socio-historical information, but they present complex problems in both internal (i.e. between the time series) and external (i.e. other datasets) interlinking [129].

The work developed by Sieber et al. [147] provides a deep analysis of how semantic heterogeneity can be addressed exclusively with semantic technologies, and describes how to achieve success in environments with very disparate data models. In the history-related domain of geographic information systems (GIS), already discussed in Section

3.4.1, Manso and Wachowicz [67] provide an extensive review on current issues in interoperability.

Classification systems

Multiple publications in classification systems [91, 98, 130, 148] are especially aimed at solving interoperability problems in historical data. Classification systems provide a standard mechanism to compare such data, but their specific implementation and effectiveness depends on the orientation towards source or goals of the historical data (see Section 3.3.2) created in phase 1 of the historical data life cycle (see Section 3.3.1).

When dealing with vast amounts of historical data, classification systems are a necessity in order to organize and make sense of the data. The main goal of a classification system is therefore to put things into meaningful groups [73]. This entails an allocation of classes which are created according to certain relations or similarities. The main issue with historical classification systems is that they are not consistent over time, making comparative historical studies problematic. Historical census data is a typical example of this problem [7, 125]. Census data is the only historical data on population characteristics which are not strongly distorted and yields an extremely valuable source of information for researchers [142].

However, major changes in the classification and coding of the different censuses, have hindered comparative historical research in both past and present efforts [138]. Researchers are forced to create their own classifications systems in order to answer their research question; however, this process often results in disparate systems, which are not comparable, contain a lot of expert knowledge, different interpretations of the data and could not be easily (re)used by other researchers. The fact that many of the modelling techniques are destructive in nature (we cannot go back to the source) makes it even more cumbersome to comprehend these sources. In order to deal with the changing classifications and vast differences at both national and international level, we need to connect the gaps between the datasets and conform to certain *standard* classification systems.

Currently several significant efforts have been made in this direction. The Integrated Public Use Microdata Series (IPUMS) project [25] for example faces the problem of bridging 8 different occupational classification systems and a total of 3200 different categories, containing the richest source of quantitative information on the American population. The North Atlantic Population Project [32] (NAPP) project provides a machine-readable database of nine censuses from several countries. The main focus of the NAPP project is to harmonize these data sets and link individuals across different censuses for longitudinal and comparative analysis. Their linking strategy involves the use of variables which do not change over time. In this process records are only checked if there is an exact match for some variables, such as race and state of birth. Other variables like age and name variables are permitted to have some variations. Another significant historical classification system is the Historical International Standard Classification of Occupations [22] (HISCO). As occupations are one of the most problematic variables in historical research, HISCO aims to overcome the problem of changing occupational terminologies over time and space. It encodes historical occupations gathered from different historical sources coming from different time pe-

riods, countries and languages, and classifies tens of thousands of occupational titles, linking these to short descriptions and images.

3.4.5 Transversal approaches

Finally, there are few but key contributions we have classified as being *transversal*, because they cover a wide spectrum of the list of overlapping tasks between the Semantic Web and historical research. They also influence almost every phase in the historical information life cycle (see Section 3.3.1).

The CLIO system [151], a databank oriented system for historians, is the first of such contributions. CLIO was, for decades, *the* system for creating, enriching, organizing and retrieving historical knowledge from historical data in the pre-Web era. Although not using Semantic Web technologies (see Section 3.2.2), it had a strong emphasis on semantics as key for structuring historical knowledge.

In the Linked Data universe, the Agora project [1] is one of such transversal contributions. It generates historical RDF of events extracted using NLP techniques from unstructured texts, uses it for enhanced search and retrieval, improves semantic heterogeneity and gives context by linking to other datasets. Similarly, the Verrijkt Koninkrijk [10] and Multilingual Access to Large Spoken Archives (NSF-ITR/MALACH) [33] projects perform these tasks in their particular domains (see Section 3.4.1). The FDR Pearl Harbor project [15] also contributes on this line, but additionally opening the very promising field of historical knowledge inference through the formalization and usage of historical OWL ontologies. All these are good examples on how historical data get much richer when their semantics are explicitly expressed and they are interlinked through standard vocabularies and ontologies.

Regarding tools, the Armadillo architecture of Semantic Web Services [2] and the Fawcett toolkit [14] contain the generic plot behind all these contributions, and cover the whole pipeline of semantic historical data management. The latter extracts RDF event-oriented triples from unstructured texts, and additionally allows historians to install a full semantic toolbox with widgets to experiment with their data. Open Refine [35], in combination with its RDF-export plugin, allows the extraction, transformation, modelling and publishing of historical Linked Data when the sources come in tabular format.

Additionally, the theoretical study by Boonstra et al. [79] envisages possibilities on how the Semantic Web can enhance research by historians. It constitutes, besides, a major work on the evolution of historical computing, ehstory and historical information science, and gives a deep intuition on how computer science can help to solve ancient problems in historical research.

3.5 Solving historical problems

In this section we point to the open historical data problems revisited in Section 3.3.3 which are addressed or solved by the Semantic Web contributions reviewed in Section 3.4 as tasks. The mapping between the open problems and the tasks is shown in Table 3.5.

Open historical data problems	Historical ontologies (3.4.1)	Linking historical data (3.4.1)	Text processing and mining (3.4.2)	Search and retrieval (3.4.3)	Classification systems (3.4.4)	Transversal approaches (3.4.5)
Historical sources (3.3.3)	✓	○				
Relationships between sources (3.3.3)		✓			✓	
Historical analysis (3.3.3)						○
Presentation of sources (3.3.3)						○

Table 3.5: Mapping between the open problems of historical data (see Section 3.3.3) and the surveyed contributions in historical Semantic Web (see Section 3.4). The sign ✓ indicates that the problem is directly addressed in the Semantic Web task. The sign ○ indicates that the problem is indirectly or partially addressed in the Semantic Web task.

The first interesting result is that some of the problems identified in *historical sources* (Section 3.3.3) are mostly solved by the approaches we review in historical ontologies (Section 3.4.1). Concretely, our perception is that the structuring of historical data and the development of historical data models have been a success due to the creation of standard vocabularies and ontologies. These ontologies aid historians to describe, at least, the baseline historical entities and relations in historical domains: events are combinations of persons, places and moments in time when something historically relevant happened. The large number of projects exposing historical Linked Data on the Web using these ontologies (see Section 3.4.1) prove their usefulness and success. There is space, though, for improvement. Although it is commonly agreed that current historical ontologies model the core semantics of historical research, authors also agree that they are still scarce and need further development [110, 136].

As part of the problems in historical sources, provision of background historical knowledge has been successful only partially. The infrastructure (Linked Data cloud, SPARQL endpoints on historical data) is set up and running. But the amount of historical data available is still too low to give good support to any historian creating historical datasets in the beginning of the life cycle (see Section 3.3.1). Consequently, little background knowledge can help today these historians in solving e.g. errors or inconsistencies at that phase. Similarly, the generic infrastructure for provenance publishing and retrieval in the Semantic Web is very mature and extensively used in other domains [160], but scarce or non existing in the historical domain although being identified as a very important requirement (see Section 3.3.3). The provision of such provenance on historical datasets needs to be guaranteed in projects using semantic technologies to publish historical data.

Solutions to the problem of *relationships between sources* are probably the greatest achievement of the application of semantic technologies to historical research. The large number of projects linking historical data we survey in Section 3.4.1 proves that the Semantic Web delivers working solutions to the problem of connecting isolated

historical data sources. The usage of developed ontologies and vocabularies has been key to this end. Additionally, the existence of classification systems (Section 3.4.4) helps on data comparability in the Semantic Web. Because we see that the body of historical knowledge in the Semantic Web is still small, we expect the problem of finding related links between historical entities and datasets to grow in the future, although the Semantic Web has generic solutions for this [146].

The problems in *historical analysis* and *presentation of sources* (see Sections 3.3.3 and 3.3.3) are only partially addressed in approaches we have classified as transversal. These works cover a wide spectrum of the life cycle of (semantic) historical data, including analysis and presentation (phases 5 and 6, Section 3.3.1). Consequently, they deal with some analyses and visualizations. However, there is a lack of contributions tackling directly the problem, or considering explicitly historical research requirements with respect to analysis and visualization. The transversal tools are hence very generic, and they could be inappropriate for some historians. Therefore, it is very important to distinguish what analysis requirements are specific to historical research, and which ones are domain-independent. Our hypothesis is that these problems overlap only partially with the goals of the Semantic Web (i.e. representing and linking meaning on the Web). However, historians could benefit from analysis and visualization tools for historical semantic data, not as specific as project-oriented, but not as generic as domain-independent.

In Table 3.5 all open problems have Semantic Web tasks providing solutions, but not all tasks are mapped to some historical open problem. Concretely, the tasks of *text processing and mining* (Section 3.4.2) and *search and retrieval* (Section 3.4.3) do not seem to solve any of the identified problems. Why do we find contributions on these areas? First, although not being identified by historians as primary problems, they constitute secondary problems that need to be solved when representing and linking semantic historical data. These problems are not exclusively historical, but they needed to be reimplemented in the Semantic Web realm (e.g. natural language processing for extracting historical RDF triples, SPARQL to query historical semantic data on the Web). Secondly, the goals these tasks aim at were quite well solved in historical research before the inception of semantic technologies (e.g. manual input of historical data, SQL queries in historical relational databases), and thus historians did not consider them into the primary problem space.

3.6 Open challenges

The use of semantic technologies has contributed significantly to solving the open problems of historical data (see Section 3.5). However, there is a lot of room for improvement. The open problems are being addressed as shown, but they are far from being solved until they get additional attention. The scarce amount of historical data on the Semantic Web is a good example. Other problems, some more specific, some more generic, could be also tackled with semantic solutions. In this section we explore some aspects of the Semantic Web that have not been used yet or could be furtherly exploited in historical research.

3.6.1 Semantics of time, change, language, uncertainty and interpretation

Classifications and ontologies in history do exist, but not for all areas, not in Semantic Web languages and not always agreed upon. Although several historical ontologies have been developed (see Section 3.4.1), these models are insufficient for the vast amount and variety of historical data that still has to be published in the Semantic Web, especially when key issues for historians like *interpretations* or *evidences* need to be modelled and conveniently linked. Historical ontologies and vocabularies have been a reality in recent approaches. Ontologies describing classes and properties of some historical concern, such as concepts around the Pearl Harbor attack in 1941 [109], are an exciting modelling exercise for researchers but also a necessary step for better structuring historical information in the Web. Ontologies and vocabularies offer a way of controlling the predicates, classes, properties and terms that the community uses as a standard for describing factual and terminological knowledge about history. Designing good ontologies for historical domains is also an area with plenty of challenges: how can ontologies comprise the many conceptions of history depending on the temporal dimension of events described [110]? Moreover, how can differences in meaning and relations between concepts be traced, as time and historical realities change these concepts [167]? To what extent these meaning differences relate to the complexity of the language (e.g. Latin, Middle languages) and uncertainty (e.g. fuzzy dates and locations)? These questions, which comprise semantic technologies, knowledge acquisition and knowledge modelling techniques, are not yet completely understood and are a significant challenge in semantic historical research. On the other side, over the centuries, dictionaries, thesaurus, classification systems have been developed. How to mount those specifically grown ordering principles to the Web in a way that makes them explorable and linkable to other ontologies is one interesting challenge which requires a close collaboration between historians, knowing and designing those specific tools, and computer scientists, often relying on much broader and generic ontologies.

3.6.2 Reasoning

From the point of view of Linked Data, ontologies and vocabularies are designed in order to control the terms in which datasets may express data, as well as the data model in which these data are represented. However, in a more Semantic Web perspective, one may expect these ontologies and vocabularies to facilitate new knowledge discovery; that is, to make explicit some implicit fact that was not trivial to deduce for the human eye, especially in big knowledge bases.

Reasoning is one of the key mechanisms of the Semantic Web still to be used in historical research. The absence of specific methods and tools for automatic historical inference, so that new, *implicit historical knowledge* can be derived, is another issue. We claim that reasoning could be fundamental for historical analysis 3.3.3 and tasks in the phase 5 (analysis) of the historical information life cycle (see Section 3.3.1).

Historical ontologies can be used to facilitate historical knowledge discovery using reasoners. Assuming that a particular domain is completely formalized as historical ontologies, then it is possible to run a reasoner on these ontologies to produce derived,

implicit rules and facts that were not present in the original model as explicit knowledge (i.e. specifically encoded in the ontology), but that were there as underlying knowledge. For instance, if an ontology describes, on the one hand, the fact that a letter was sent from one government to another, and on the other hand, the fact that governments have a person responsible of sending and receiving letters, then it may be possible for the reasoner to infer what concrete persons sent and received what letters. As the knowledge base grows, implicit knowledge is not evident anymore and reasoners can facilitate an enormous work and produce high-value pieces of historical knowledge.

Since historians have different interpretations and no clear research question when starting an investigation, abductive reasoning (i.e. given the conclusions and a rule, try to select possible premises that support the conclusion) may be more convenient than deductive reasoning (i.e. deduce true conclusions given a premise and a rule) in historical research [84, 107]. These would revert the order of some phases of the life cycle of historical information (see Section 3.3.1), generating a more bottom-up, data-based generation of hypotheses supporting evidence. The impact of abductive reasoning in historical research and its relationship with the life cycle needs further study and clarification.

The introduction of any kind of reasoning in the life cycle needs to be done with the goal of supporting, not replacing, the task of the historian, who must keep control of the implementation of the different phases.

3.6.3 Linking more historical data

We show in Section 3.4.1 that great efforts are being devoted to publish historical Linked Data. However, the amount of structured historical knowledge available on the Web is still insufficient to aid tasks that need high amounts and different kinds of historical background knowledge. While many different data and information sources exist, they are not always interlinked. This isolation of historical data sources hampers that they can be found, but it also inhibits how they can be further processed and connected.

One of the big claims of linked data is that, by linking datasets, relations established between nodes of these datasets highly enrich the information contained in them. That way, browsing datasets is not an isolated task anymore: by allowing users (and machines) to explore entities through their predicate links, data get new meanings, uncountable contexts and useful perspectives for historians.

For example, consider a scenario with three different SPARQL endpoints exposing RDF triples of a census with occupational data, a historical register of labour strikes, and a generic classification system for occupations (in the context of one particular country, for instance). Suppose that: the occupational census of the data exposes triples with countings on occupations (for example, how many men and women worked in a particular occupation in a concrete city), the historical register of labour strikes contains countings on how many people participated in labour strikes (number of women and men, per occupation and city), and the generic classification system harmonizes names of the occupations between both previous datasets (for example, gives a common number for representing occupation names that may vary between census occupations and labour strike occupations). Then, it is clear that several SPARQL queries

can be constructed to give very meaningful and interesting linked data to the historian. For instance, such a query may return, given a city and an occupation code, which ratio of men and women followed a particular well-known labour strike. Another SPARQL query may return an ordered list of historical labour strikes by relevance, according to several indicators (strike successfulness ratio, total number of workers on strike, density of people on strike depending on the location, etc.). It is obvious that the possibilities increase if we think of more related historical sources to link, like datasets describing historical weather or historical geographical names and areas.

3.6.4 Flexibility of data models

It is considered to be a bad practice in historical research not doing the historical data modelling at phase 1 of the historical information life cycle (see Section 3.3.1). The choice of a particular data model to represent historical data is a critical issue for most historical computing projects. The election of some appropriate data model may seem a good design decision at some stage of the project. However, new requirements, research directions or stakeholder priorities may convert that data model into an obstacle more than an aid. Flexibility of data with respect to the data model used to represent historical domains is desired to avoid restructuring entire databases. Comparison in historical research requires flexibility of the models to be able to match them to one another. At the end, that enforces historians to make their data selection and processing dependent of a certain data model that can not be easily replaced or altered if needed. This happens usually in environments with changeable and creep requirements [111].

Applying semantic technologies and Linked Data principles to historical data may have a major advantage regarding historical data models, providing flexibility at the historical data modelling phase. Two different approaches regarding historical data modelling have been followed traditionally in historical computing: the *source-oriented* representation, and the *model-oriented* (also known as *goal-oriented*) representation (see Section 3.3.2). Do semantic technologies allow a flexible representation of historical data? Can the Semantic Web found a new standard on a source-and-goal, hybrid approach? RDF databases (see Section 3.2.2) can store the middleware representation of further views on the data [125]. These views can be modelled as close to any particular historical interpretation as needed. This way, the decision of what data model suits better the historical source can be postponed until the very end of the life cycle (see Section 3.3.1), or adopted as early as necessary.

Moreover, additional questions arise when considering the traditional perspectives on data modelling: the conceptual, logical and physical data models. These perspectives help in detaching data management technology, like relational databases or RDF triplestores, from conceptual schemas (i.e. the semantics of a domain). While conceptual data models are currently shared on the Web as e.g. historical ontologies (see Section 3.4.1), the flexibility of the whole modelling stack towards semantic changes needs to be better understood.

3.6.5 Non-destructive data transformations

The non-flexibility of data models (see Section 3.6.4) is related to the non-flexibility of historical data transformations. Historical data are modified in the life cycle of historical information (see Section 3.3.1). But if update, enrichment, analytic and interpretative operations are not controlled, these transformations lead to different historical data representations which can hardly be related to each other any more, nor in terms of provenance nor in terms of relatedness.

Another issue is supporting data transformations under two constraints: (a) without modifying source data (so the originals stay intact); and (b) with tracking of changes. Consequently, destructive updates are a major concern when selecting, aggregating and modifying historical data. On the one hand, modifications to specific encodings (CSV, spreadsheets, XML) do not support non-destructive updates, and version control systems are necessary to retrieve previous states. On the other hand, relational databases can be inefficient when querying all recorded transformations, edits and manipulations.

Non-destructive updates are well supported by current Semantic Web technology like SPARQL (see Section 3.2.2). SPARQL `CONSTRUCT` allows the construction of RDF triples according to the supplied graph pattern, facilitating data transformations without altering consistency of previous states in the knowledge base. SPARQL `SELECT` selects, according to some graph pattern, the desired data in the RDF graph, and disposes them according to any desired view format (for example, columns matching some interesting historical variables). However, standardization on how semantic technologies cover data transformations in all phases of the historical data life cycle (see Section 3.3.1) is needed.

3.7 Discussion and conclusions

In this paper we present a general overview of semantic technologies applied to historical research. We describe a general approach to historical research and the Semantic Web, and motivate why the combination of the two is an interesting field of research. We introduce core elements of historical research, such as the life cycle of historical information, several classifications for historical data, and the open problems shared by historians and computer scientists. Then, we overview contributions to the young historical Semantic Web in form of papers, projects and tools, articulating the work into several tasks and trends within these tasks. We provide a mapping to see to what extent the work on these tasks is helping to solve the open problems of historical data and historical research. Finally, we dig out a list of interesting open challenges for the future, like working out the semantics of critical aspects for historians, such as interpretation and time, and encouraging reasoning in the historical Semantic Web.

It is interesting to observe the sparsity in Tables 3.4, 3.1, 3.2 and 3.3. There is a significant difference in the number of empty spaces (i.e. specificity of the contributions) between Tables 3.4 and 3.3 (papers and tools, ontologies), and Tables 3.1 and 3.2 (projects and online resources). While the former set has essentially lots of *holes*, the latter has lots of *complete lines*. The reason for this is probably the specificity researchers think research papers and useful tools need. Usually written by computer

scientists, papers and tools need to be grounded and tackle a very concrete problem to be worth written or implemented. On the other hand, projects (Table 3.1) are written in a very generic way covering all tasks, with probably intensive participation of historians and clear aims to solve the whole pipeline. In practice, though, these goals are materialized in very concrete research contributions. This leads us to think that Semantic Web solutions need very specific requirements in order to be correctly deployed in history. They need to be applied to historical data in a complex, layered and properly adapted pipeline. Good practices and standards, and their relationship with the life cycle of historical information, are still needed for the field to continue evolving.

We show how the Semantic Web and history communities understand the need for representing inner semantics implicitly contained in historical sources, and how these semantics can be conveniently identified, formalized and linked. With the appropriate pipelines, algorithms can extract entities from digital historical sources and transform these occurrences into RDF triples, according to some historical ontology or vocabulary. These entities can be linked between them and with other historical Linked Data, contributing to an open, world wide, online persistent graph of historical knowledge: an historical Semantic Web. The work presented in this survey contributes in one phase or another in this graph-building pipeline. We leave to the reader if this historical Semantic Web building pipeline is, in fact, the Semantic Web version of the life cycle of historical information.

The challenge of the realisation of a historical Semantic Web meeting as many requirements as possible may bring new facilities for a number of stakeholders. On the one hand, humanities researchers, also outside history, will be able to integrate the historical Semantic Web to their own information life cycle. They will be able to search, retrieve and compare historical knowledge and use it for the construction of their narratives, still the final outcome of historical research. On the other hand, practitioners will be able to search new data sources to develop history-aware applications for public institutions, private companies and citizens.

Chapter 4

What Is Linked Historical Data?

4.1 Historical Sources as RDF Graphs

Historical sources have traditionally been encoded in different formats: from papyrus to digital images, through books, tapes and photographs. It is not difficult to see the benefits of publishing historical sources as Linked Open Data [126]. However, it is unclear whether standard Linked Data modeling and publication pipelines are suitable for historical sources. In this paper, we are interested in modeling the fundamental properties of historical sources in order to make explicit to what extent current Linked Data publication procedures are adequate.

Independence and *reliability* of sources are fundamental issues historians take into account in scholarly writing [106]. To address these, historians distinguish between *primary* and *secondary* sources. Primary sources are “original materials created at the time under study that have not been altered or distorted in any way” [?, ?]. Secondary sources are “documents that relate or discuss information originally presented elsewhere, written after the fact with the benefit of hindsight” [?]. A fundamental difference between the two is that primary sources must be *immutable*: they cannot be altered once they are created. Traditionally, immutability of sources is achieved through *archiving* them, either as books (in a library or book archive), as physical objects (in a museum archive), or more recently as digital objects (in a digital archive). The archive is the authority that protects the primary source from change, providing *independence* and *reliability*. As a consequence, primary sources are inevitably detached from their original context. Secondary sources are attempts from historians to recreate this context.

A strict requirement then is that RDF graphs of primary sources need to be immutable as well. But how does RDF deal with change over time [171]?

Intuitively speaking, changes in the universe of discourse can be reflected in the following ways:

1. An IRI, once minted, should never change its intended referent.

2. Literals, by design, are constants and never change their value.
3. A relationship that holds between two resources at one time may not hold at another time.
4. RDF sources may change their state over time. That is, they may provide different RDF graphs at different times.
5. Some RDF sources may, however, be immutable snapshots of another RDF source, archiving its state at some point in time.

Statement 1 is problematic: a primary source that changes keeps its IRI although its identity is changed (see Section 4.2). In addition, statements 4 and 5 have important consequences for historical sources. First, it follows the *alive-dead* Linked Data dichotomy: on the one hand, there is a *living* LOD cloud that is constantly updated and changed; on the other hand, a *dead* and archived LOD cloud exists as old snapshots of what once was alive. This situation corresponds to the life cycle of primary and secondary sources. All sources are first ordinary living LOD data, but the fact that they are *archived to preserve their immutability* turns them into primary sources. The metaphor of the *alive-dead* LOD serves well the purpose of primary and secondary sources as RDF graphs. For a primary source to be represented as an RDF graph, it is necessary (and sufficient) to be archived and preserved from change. RDF graphs of secondary sources, on the other hand, live in the LOD cloud similar to other datasets.

4.2 An Ontological Framework for Historical Sources

What are the basic ontological properties that characterize historical sources? In order to come with appropriate proposals on how to publish historical primary and secondary sources as LOD, we first need to understand their fundamental characteristics. We apply the philosophical stances of *perdurantism* and *endurantism* and the OntoClean methodology of [101] to study ontological properties that apply to historical sources; we model these sources according to their properties.

Perdurantism holds that ordinary things like animals, boats and planets have temporal parts (things persist by *perdurating* through time). *Endurantism* is the stance that ordinary things do not have temporal parts; instead, things are wholly present whenever they exist (things persist by *enduring*) [113]. The DOLCE ontology [97] translates these stances to two types of entities: *endurants* and *perdurants*, which can be characterized by whether or not they can exhibit change in time. Endurants “can “genuinely” change in time, in the sense that the very same endurant as a whole can have incompatible properties at different times; perdurants cannot change in this sense, since none of their parts maintain identity in time.” Secondary sources, such as comments, notes, articles, annotations, are *endurants*; at any point in time they can be appreciated as a whole, while they still may undergo changes (e.g. working papers). Primary sources, on the other hand, have the same enduring properties, but *cannot* change: if any of their properties change, they lose identity. The accumulation over time of secondary sources that share a dependency on a primary source form our “body of knowledge” about the historical entity represented by the primary source. This accumulation is a **perdurant** (similar to item 4 in Section 4.1). On the other hand, the “state of our knowledge” at any point in time is a *slice* of that body of knowledge: a collection/set of **endurants**.

Distinguishing between perdurants and endurants is closely related to the question of identity: if sources can change over time, how can we guarantee that they are the same entity? To help answering this question, we use the OntoClean methodology [101]. Following OntoClean, some properties are essential to *all* their instances; we call these properties **rigid (+R)**. For instance, an entity having the property of *being a person* is guaranteed to preserve its identity even if other of his properties change, because *being a person* is rigid. Properties that are not essential for *some* of their instances (i.e. required to change) are called **non-rigid (-R)**; of these, properties that are not essential to *all* of their instances (i.e. required to change) are called **anti-rigid ($\sim R$)**. According to this, the property of a source being primary is **+R**, because being primary is essential to all sources (i.e. if it stops being a primary source, it no longer exists); the property of being secondary is $\sim R$, since secondary sources may become primary sources through archiving.

Another way of looking at identity of historical sources consists of considering them as *sortals*. A **sortal (+I)** is a class all of whose instances are identified in the same way. The class of secondary sources does not carry any identity criteria (i.e. a secondary source cannot be identified by any predefined set of characteristics). On the other hand, a primary source is always **+I**: its identity criteria cover all of its properties (in order for something to be a primary source, none of its properties is allowed to change).

Unity (+U) is the metaproperty of classes all of whose individuals are *wholes* under the same relation. A *whole* is an instance that, in opposition of *mere sums*, does not create new instances of the same class it belongs to when an arbitrary subsection of such instance is considered. For instance, splitting a piece of *clay* in two, constitutes two pieces of *clay* (it is a *mere sum*), while this does not typically happen with e.g. instances of the class *person* (a *whole*). Primary sources are **anti-unity ($\sim U$)**, since any part in which a primary source may decompose creates a new primary source. Think of historical objects for which only certain parts could be preserved; these parts constitute the genuine primary source. In case new parts of the object were found, these would constitute different independent primary sources. Secondary sources are **+U** because specific relations between their parts preserve their integrity as wholes, and arbitrary parts of them do not constitute secondary sources anymore.

Finally, a property is **dependent (+D)** if each instance of it implies the existence of another entity [101]. Primary sources are **independent (-D)**, given that they can exist independently of other entities. However, secondary sources are **+D**: every secondary source is always *about* some existing primary source.

Table 4.1 shows the correspondence between the properties of primary and secondary sources. We model historical sources using the study of this Section and the considerations made in Section 4.1.

Secondary sources	Primary sources
Endurant	(Strong) Endurant
Alive datasets	Dead datasets
Non-timestamped resources	Timestamped resources
Dereferenceable IRIs	Archive-only-dereferenceable IRIs
Anti-rigid $\sim\mathbf{R}$	Rigid $+\mathbf{R}$
Dependent (on the primary Source) $+\mathbf{D}$	Independent $-\mathbf{D}$
Not a sortal $-\mathbf{I}$	A sortal $+\mathbf{I}$
Unity $+\mathbf{U}$	Anti-unity $\sim\mathbf{U}$

Table 4.1: Ontological metaproperties of historical sources.

4.3 Linked Historical Data: From Modeling to Publishing

The model proposed in Section 4.2 conflicts with some of the basic LOD publishing principles, more concretely with the openness of the Web. The AAA rule (Anyone can say Anything about Any topic) is one of the essential principles of the Web, which also holds for RDF data. The IRI of a primary source can be used by anyone as a subject of an RDF statement; this changes the graph of the primary source, and breaks the basic principle of immutability of primary sources (see Section 4.1). In this Section we investigate mechanisms to publish Linked Historical Data as LOD according with the model of Section 4.2.

To solve this conflict, we propose *dereferenceability* as a mechanism to preserve the fundamental properties of primary sources and their interplay with secondary sources. Concretely, we propose a dereferencing service for authoritative digital archives hosting RDF graphs with two essential characteristics: *reliability* and *independence* (the overlap with the requirements for historical sources in Section 4.1 is no coincidence). First, dereferenceability is the *only* mechanism by which users may know that they are talking about a primary source. Hence, it is necessary that, when asked, the authoritative archive provides information on whether it *knows* something about such primary source or not (e.g. via SPARQL ASK queries). This way we achieve *reliability*: the only reliable primary source triples are those for which the archive returns valid descriptions. This will not happen with triples about the primary source issued by anybody else.

Second, when users dereference triples of a primary source, they get back *copies* of it, where all IRIs are replaced by new ones, but refer back to the original primary source IRI (or IRIs inside the primary source graph) through `prov:wasDerivedFrom` relations. This way we achieve *independence*: new statements (i.e. secondary sources) refer to this qualified copy independently on the original contents of the in-archive primary source graph, preserving its independence and immutability. Alternatively, the resolution creates a new FRBR expression of the same FRBR work, but then the primary source is also an expression (the first). URN-like tricks (cf. DOIs) could also be used, such that users have to use a trusted dereferencing service at the archive location to obtain the primary source data (e.g. crossref dereferences DOIs at <http://dx.doi.org/> to the printer’s page, or to an RDF representation of the source).

With our proposed model and publishing study, we can now answer the question: what is Linked Historical Data? Historical data is the union of *primary sources* P and *secondary sources* S : (a) a *primary source* P is an accumulation of strong-endurant, dead, timestamped, only-archive-dereferenceable, rigid, independent, sortal, and anti-unity resources; (b) a *secondary source* S is an accumulation of endurant, alive, non-timestamped, dereferenceable-by-anyone, anti-rigid, dependent, non-sortal, and unity resources; (c) statements in S contain links to statements in P ; and (d) for any time t in which a statement of P is made, and for any time t' in which a statement of S is made, $t' > t$.

In this paper we argue for an ontological model and a consequent adequate publication of historical sources as RDF graphs in the LOD cloud. We advocate the use of the OntoClean methodology and DOLCE to give characterizations of *primary* and *secondary sources*. We propose the implementation of specific IRI dereferencing services in digital archives to preserve these fundamental properties of historical sources in the form of *independence* and *reliability*.

Part II

**Integration of Linked Statistical
Data**

Chapter 5

LSD Dimensions

Chapter 6

From Flat Lists to Taxonomies: Bottom-up Concept Scheme Generation in Linked Statistical Data

6.1 Introduction

Motivation. RDF Data Cube is the standard for publishing multidimensional data in the Semantic Web, allowing linkage to other concepts and datasets in the so-called Linked Statistical Data (LSD) cloud. Statistical datasets often come in non-standardized ways, making it difficult to deal with comparability: a variety of dimension and value choices make it necessary to clean, correct and standardize the data before working with it. A common standardization practice is the creation and use of concept schemes. Some current concept schemes standardize common statistical concepts [10]. However, the lack of standard concept schemes in other domains is a great bottleneck for LSD publishers that wish to leverage (and ensure) reusability of concept scheme knowledge. Consequently, users are confronted with manual procedures in order to put dimension values (or codes) into meaningful groups.

Problem statement. Currently available concept schemes are not sufficient in order to standardize LSD. Researchers have to deal with unstructured and non-standardized dimension values, and are moreover confronted with ‘information deluge’, making the process of putting things into meaningful groups extremely complicated and time consuming, even with expert knowledge. Tools aiding this concept scheme building process for already published LSD are highly needed and non-existent, hampering the comparability and use of statistical datasets across the Web.

Use-cases. To enhance comparability studies in social history, researchers have been studying the dimension values of several historical LSD datasets. They have great interest in proposing concept schemes for the standardization of historical reli-

gious denominations and historical housing types.

Contribution. We propose a highly reproducible, generalizable and scalable workflow to automatically generate standard concept schemes from non-standardized dimension values in LSD datasets in a bottom-up way. We leverage the intrinsic lexical and semantic properties to propose meaningful classifications.

Findings. We find that the combination of lexical hierarchical clustering and semantic tagging of non-standard dimension values of our workflow provides useful support to the knowledge expert in the concept scheme building process.

The rest of the paper is organized as follows. We survey the related work in Section 2. In Section 3 we propose a workflow to automatically construct concept schemes from flat literals of non-standard values of dimensions in LSD. In Section 4 we present our experiments in Linked Census Data, before we conclude in Section 5.

6.2 Related Work

When working with non-standardized statistical data, the process of creating classification systems has been a mostly manual job. Current classification practices are therefore based mainly on data-driven, bottom-up, manual efforts by domain experts [1]. Researchers which lack programming skills, budget or sometimes necessitated by the data itself are bound to use (a combination of) different tools in order to clean, filter, group and classify statistical data before its publication: this is the purpose of OpenRefine [7]. A set of clustering algorithms (defined as “finding groups of different values that might be alternative representations of the same thing”) are provided. Perhaps [8] is the closest match to the taxonomical knowledge construction via hierarchical clustering that we aim at, although fundamental differences apply with respect to the input data (collections of documents instead of flat literal lists) of different domains. Unfortunately, there is hardly any tool support available for conducting this standardization: (a) in a purely Linked Data setting; and (b) standardizing values after their publication as LOD in order to preserve both original and standard values.

6.3 Bottom-up Construction of Concept Schemes

We propose a workflow to automatically build bottom-up concept schemes from flat lists of non-standardized dimension values in LSD. We aim at RDF Data Cubes that need to preserve faithful representations of original source data: in such datasets, it is not possible to standardize dimension values before converting to RDF Data Cubes. The process is divided in five steps: retrieval of literals, hierarchical clustering, semantic tagging, linking and serializing.

6.3.1 Retrieval of Literals

First, literals of the non-standardized dimension values need to be retrieved. Since we are interested in building concept schemes in LSD, we use SPARQL queries that follow

the template shown in Figure 1 against RDF Data Cube [4] datasets. Once executed, the resultset contains a list of unique non-standard dimension value literals.

```
PREFIX qb: <http://purl.org/linked-data/cube> PREFIX skos: http://www.w3.org/2004/02/skos/core
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema>
```

```
SELECT DISTINCT ?dimValue ?literal WHERE ?obs a qb:Observation; eg:my-
dim ?dimValue . ?dimValue skos:prefLabelrdfs:label ?literal .
```

Fig. 1. SPARQL template to retrieve non-standardized dimension value literals in RDF Data Cube datasets. eg:my-dim represents any entity of type qb:DimensionProperty.

6.3.2 Hierarchical Clustering

Our hypothesis is that knowledge experts group disparate literals mostly on a string similarity basis. Obviously, some literals may be grouped together for other reasons (e.g. semantic similarity), and it is part of our study to understand which ratio of the target concept scheme can be reached using lexical criteria only.

Since concept schemes are taxonomies, we choose hierarchical clustering as our method to build taxonomic relations between non-standard literals. We use the resultset of the previous step as input for the hierarchical clustering algorithm included in SciPy [3], and we use the Levenshtein edit distance [2] as a distance metric.

6.3.3 Semantic Tagging

An important task knowledge experts do when they build concept schemes is to label upper categories (e.g. the cluster containing “Barracks”, “Arsenal”, and “Citadel” may be named “Military buildings”). We suggest meaningful names for the output clusters of the previous step by leveraging semantic resources like WordNet and DBpedia. Concretely, we offer two alternatives for semantic tagging of clusters:

1. Term-based tagging. After the removal of stop words, we tokenize and stem all literals under the same cluster and rank them according to their appearance frequency. We use the top-1 token to query WordNet and DBpedia to get all of its synset and skos:Concept, respectively. We use those as suggestions to name the cluster.
2. Bag-of-words tagging. After the removal of stop words, we tokenize and stem all literals under the same cluster. We query WordNet and DBpedia using all tokens of all literals of the cluster, getting their synset and skos:Concept. We leverage skos:broader relations to find the closest common broader concept of all literals, and we use this concept as suggestion to name the cluster. We consider all the descendant links below a cluster node k to belong to the same cluster if k is the first node below the cut threshold t . We use $t = 0.7 * \max(d(k, i))$, where $d(k, i)$ is the distance between the node k and any other node i .

6.3.4 Linking

After producing the concept scheme, we still need to link it to the original non-standard values. Since we have preserved the URIs of the original dimension values (see Figure 1), issuing links between the two is an almost trivial task.

6.3.5 Serializing

Once we have produced the concept scheme and the links back to the original dimension values, we serialize both datasets using SKOS [5] and RDF Data Cube [4], producing URIs for all new concepts.

6.4 Experiments with Linked Census Data

We use the workflow proposed in Section 3 to build bottom-up classifications of non-standard dimension values in the RDF Data Cube version of the Dutch Historical Censuses dataset (CEDAR) . This dataset is produced by TabLinker , converting Excel census tables to RDF Data Cube [6]. We use our approach to build classification schemes on top of non-standard dimension values in this dataset.

6.4.1 Input Data

The CEDAR dataset covers a long time period (1795-1971) in which lots of religious denominations were registered in a non-standard way. We aim at producing a standard concept scheme to cover all these religious denominations . Similarly, the census also registered counts for houses, encoding a dimension house type in a non-standard way. We aim at producing a standard concept scheme to cover all housing types . Our gold standards are classification schemes developed by knowledge experts on top of the Dutch historical censuses. For historical religions, researchers manually standardized and coded the variables belonging to the same religious denomination . The final outcome is a classification system of historical religions containing 210 unique denominations. For historical house types, we use another expert-based classification based on a manual and straightforward approach in which the terms are classified according to their functions . We compare the results of our proposed workflow with these expert-crafted classifications.

6.4.2 Results and Discussion

We execute several times our workflow on the input datasets, exploring appropriate parameter values for hierarchical clustering. We take the average term

distance to determine the distance between clusters. Figure 2 shows our resulting schemes.

Fig. 2. Dendograms of the hierarchical clusters of CEDAR house types and historical religions. See <http://goo.gl/ggMqiR> and <http://goo.gl/VG9IGU> for full resolution images.

We observe interesting groups being identified in the building types dataset. For instance, the cluster containing the values “Klooster der Franciscanen”, “Klooster van de orde der Franciscanessen” and “Klooster van de orde der Benedictijnen” clearly identifies kloosters (monasteries), and gets appropriately <http://nl.dbpedia.org/resource/Klooster> (and its English equivalent <http://dbpedia.org/resource/Monastery>) as a semantic tag for the broader category of the concept scheme. Similarly, instances of historical religions that identify “Apostolic” or “Protestant” denominations are grouped together under the same cluster. Interestingly, a purely lexical approach exploits the transitivity of some string similarities (e.g. “Kazerne” and “Militair Ziekenhuis” are clustered together due to the linking member “Militair Kazerne” of the same cluster). On the other hand, the purely lexical clustering shows also its limitations when instances like “ziekenhuis” (hospital), “armhuis” (poorhouse) or “weeshuis” (orphanage) are clustered together (due to their common suffix “-huis”) despite their notable semantic differences.

Knowledge experts validating our workflow compare these results with the gold standards, and see its usefulness when building concept schemes to standardize historical statistical data. Concretely, they are interested in its application as a knowledge support tool in the concept scheme building process. Accordingly, a key issue of the process, covered by our workflow, is leveraging the combination of lexical and semantic structuring. Experts truly think that a combination of both approaches is what indeed goes on when they execute the process manually. It is to be seen, though, the trustworthiness of our proposed workflow as a totally autonomous tool.

6.5 Conclusions and Future Work

In this paper we present an automatic approach to generate concept schemes from non-standard dimension values in Linked Statistical Data. We propose a workflow that combines hierarchical clustering to leverage lexical relatedness, with the enrichment from external knowledge bases to leverage semantic relatedness. As a result, we produce concept schemes that knowledge experts can compare with their manually generated ones. We plan to extend this work in multiple ways. First, we will systematically compare our workflow output with the gold standards, in order to get precision/recall scores that evaluate our approach. Second, we will execute the workflow against arbitrary datasets, to confirm its domain independence. Third, we plan on finding optimal values of the threshold, here set by empirical exploration. Finally, we will generalize our proposal by implementing additional clustering algorithms (e.g. Latent Semantic Analysis) and other semantic methods for cluster tagging.

6.6 References

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Chapter 7

Linked Humanities Data: The Next Frontier? A Case-study in Historical Census Data

7.1 Introduction

Census data plays an invaluable role in the historical study of society. In the Netherlands, Dutch census data are among the most frequently consulted sources of statistics of the Netherlands Central Bureau of Statistics [1]. The Dutch census data set is unique in that it covers a period of almost two centuries (1795- 1971). During that period, it is the only regular statistical population study performed by the Dutch government. For the period before the 20th century, the census is the only historical data on population characteristics that is not strongly distorted [12]. Census data is both large and complex [1,9,8]. Collected over centuries, it is a rich source of scientific insights into social history. Qualitative information about the development of professions, habitation and religion are up for grabs: if only they were accessible in a uniform, consistent and systematic way. Efforts on census analysis are hampered by the heterogeneity of the data. Data is not only available in different formats, but internally it uses different classification systems, metadata schemes, inconsistent (and incorrect) representations, temporal invariance, various variable coding schemes or differences in the semantics of annotations. Comparative studies are unfeasible in this context. To solve this, social historians follow a (manual) process of harmonization: the creation of a unified, consistent data series from disparate census samples [2].

Linked Data appears to be a natural fit, and we decided to publish harmonized census data as RDF. RDF offers a uniform, schema-agnostic representation format that allows a transparent mapping across censuses, while maintaining a link to the original source. However, social scientists and historians require that the process by which the original source data is transformed into the harmonized data remains explicit: the original

data and intermediate steps must always be traceable from the result. Therefore, we choose to use RDF to represent census data faithfully, without committing to a particular conceptualization and applying Linked Data principles from the very start of the harmonization process. We analyze how, when and where data problems arise; and study how harmonization practices can be generalized and applied in a Linked Data context.

This paper contributes a description of the use of Linked Data to represent and harmonize statistical data in the socio-historical domain, describing a harmonization methodology based on standard Linked Data principles. We identify the gaps in this approach, and illustrate how the size and complexity of the resulting graphs poses new challenges for Semantic Web technology. The paper is structured as follows. Section 2 describes the Dutch census dataset in more detail, introduces its problems and summarizes how social historians solve them. Section 3 introduces the RDF model for linked census data, how we generated it from the original data, and how we evaluated its correctness. In section 4 we describe the problems we encountered applying standard Linked Data practices to the census due to its size and complexity, illustrate new challenges for Semantic Web technology to tackle these problems, and discuss potential solutions.

7.2 Census data and harmonization

The Dutch historical census dataset comprises 507 Excel workbooks containing 2,288 tables, being a digital representation of the original census books. These books have been digitized as images and manually translated into Excel workbooks. A version of the dataset is available at <http://www.volkstellingen.nl>.

One Excel workbook may contain several tables (one per sheet), but given an Excel workbook it applies only to one year and one census type (population census, occupation census or housing census). Some years (1795-1879) only have the population census, but all three types of censuses are present from 1889 onwards. The last censuses of the 19th century contain a higher number of tables because the complexity of these censuses, according to experts (see Figure 1).

The dataset contains 33,283 annotations at the cell level, with a similar distribution to the tables (see Figure 1). Annotations contain considerations (e.g. this cell includes 2 persons of unknown age, this number is unreadable), interpretations (e.g. this total should take foreigners into account) and corrections (e.g. 48 instead of 43) made by digitizers, archivists or social historians.

We observe structural heterogeneity while comparing these tables (see Figure 2): row and column headers do not follow any recognizable pattern with respect

Fig. 1: Dutch historical census dataset counts for tables and annotations.

to their contents or sorting. Sometimes contents of row and column headers are denoted hierarchically, spanning through several columns or rows. Other issues, like unstructured subtotals in the data cells, notes written outside of the table borders and un-

structured provenance in annotations' body make it even harder to identify a common structure. Our goal is to solve these differences, so all tables can be uniformly queried.

7.2.1 Harmonization semantics

We observe also semantic heterogeneity in these tables: meanings and values of variables (in the statistical sense) are not compatible in cross-year comparisons. We describe the main topics where semantic discordance plays a role, identifying specific data issues, and additional requirements.

Occupation is one of the most unwieldy variables in censuses. Comparative historical research on occupations is heavily hampered by misunderstandings with regards to occupational terminology across time and space [8]. Throughout the years occupational categories of censuses worldwide have evolved significantly, emerging and disappearing due to occupation evolution, and thus changing classification systems [11]. Accordingly, this evolution is reflected in the fact that the U.S. Census listed 287 separate occupations in 1950, compared to 441 in 1971. We have occupations such as tile settlers and roof repairers which were sometimes presented as one occupation and in other years separated.

Demographic information is at the core of the census. The composition of age distribution across different census years is a challenging aspect. Dissimilar age ranges are applied throughout different censuses making historical comparisons cumbersome.

Fig. 2: One of the census tables. Legend illustrates common groups of cells identified.

The census contains rich geographical coverage of the Netherlands' state and spatial divisions. Giving geographical context to data enhances its meaning and makes allowances for their environment so as to enhance their comprehension [13]. However, comparative studies are severely hampered by changing boundaries. The composition of the municipality of Rotterdam, e.g., has changed significantly over time. Smaller municipalities may not exist anymore or have merged. This makes it difficult to trace these back in history without a historical classification system and GIS (geographic information system) referencing.

Errors have been introduced at many stages of the tables cycle: during its collection, during the digitization process, or in translating from images to Excel workbooks. Since we have to keep these workbooks faithful to originals, erroneous numbers can not simply be overwritten with corrected ones.

Provenance on activities that imply accessing, rewriting, reviewing or transforming the contents of the tables needs to be recorded. Who did what, when, why? are significant questions for researchers using these tables.

7.2.2 Current harmonization practice

We talked extensively with social historians to understand harmonization, the process of creating a unified, consistent data series from disparate census samples [2]. The cur-

rent harmonization practice mainly uses external classification systems such as HISCO to link and standardize the different occupations. By an

The HISCO code is used to study historical patterns of work and social mobility in comparative perspective, and is built upon the 1000 most frequent male and female occupational titles (see <http://hisco.antenna.nl/>.)

extended process of trial, error and review a set of coding rules was developed allowing titles to be incorporated into the classification scheme. Compatibility with the International Labour Organisation's ISCO686 scheme allows the linkage of historical and contemporary data sets.

The current practice consists of a straightforward process of mapping occupation codes into an external classification system, such as HISCO. In most cases users can manually check the index for a given occupation title and find the corresponding HISCO code. In more difficult cases users have to choose themselves between different codes or in cases where the title cannot be found to proceed by using the logical structure of the HISCO code using the major, minor and unit structure. This also applies when harmonizing municipalities. As the internal classifications differ too much, several external classification systems (Amsterdam Code, CBS Code, Wageningen Code etc.) have been developed applying the principle of maximum and minimum varying codes [10].

The census applies disparate age distributions throughout the tables, making comparisons difficult. Harmonization uses statistical aggregation and imputation for solving this. For example, comparison of age ranges 14-18 and 19-20 with 14-15 and 16-20 can be achieved with the aggregate 14-20, or by way of interpolation dening 14-15, 16-17 and 18-20.

7.3 A structured approach on harmonization

Since harmonization is the process social historians use to resolve data issues, it seems that the whole dataset should be harmonized before applying Linked Data principles. This approach has been followed by other initiatives, such as the United States 2000 census and the 2001 Spanish census have published census data as RDF, but in a microdata-based approach (microdata considers individual data at the municipality level, whilst census data consists of aggregated data). This seems the more straightforward approach, but there are several reasons to do the opposite: that is, to produce an RDF version of the dataset, and harmonize afterwards.

The first reason for this is that harmonization is not a standard process. Harmonization strongly relies on interpretation and specific goals. Researchers should be free to make their own interpretations and establish their own goals on how to harmonize census data. This is what historical research is all about.

Secondly, access to source data must always be guaranteed. Original and harmonized data must coexist in a single system to fulfill requirements of both source-oriented and goal-oriented historical research methods

Finally, as we show in section 4, Linked Data principles and semantic technologies can play an important role in solving and implementing harmonization.

⁶ See <http://www.rdfabout.com/demo/census/>, only one year census, only microdata.

⁷ See http://gutenberg.dcs.fi.uva.es/bhscmcyt/census/sparql_en.php, *onlyoneyearcensus*, *onlymicrodata*.

In a source-oriented approach, historians have a priority on representing research data with a faithful schema with respect to structure of their sources. In a goal-oriented approach, the priority is to represent these data following a particular conceptualization of the domain that suits functional requirements. [1].

We have designed and implemented a faithful conversion of the Dutch historical census data from Excel workbooks to RDF. After designing appropriate data model and vocabularies, we developed a script producing RDF graph data, TabLinker, and a number of SPARQL queries to validate our requirements.

7.3.1 Generating census triples

We developed a specific tool, TabLinker, to produce RDF graphs according to this data model. Despite the existence of other tools translating tabular data into RDF (like csv2rdf4lod or TopBraid Composer), some of them used in the US census case, none met our input data (e.g. disparate table structure), output data (e.g. annotations, provenance support) and, most importantly, harmonization requirements. We organize all produced triples by TabLinker in a three layer architecture.

First, the source layer consists of a 1:1 copy of table contents. It represents tables in a flat, schema-agnostic way, providing direct access to original data. We use RDF to avoid an early, concrete conceptualization of the domain, according to principles of historical source-oriented approaches. Any necessary transformation due to functional requirements has to be done in a further stage (e.g. via CONSTRUCT SPARQL queries).

Since data contained in a census table is statistical data, we designed a per-cell data model according to the RDF Data Cube vocabulary (see Figure 3). We use the term observation for all metadata describing a data cell and its context (d2s:isObservation a qb:Observation). A data cell is affected by a number of dimensions (d2s:dimension a qb:DimensionProperty) that correspond to the column and row headers that apply to that cell. The content of a data cell is always a number counting population (d2s:populationSize a qb:MeasureProperty).

We use the Open Annotation Core Data Model and PROV to add annotations and provenance to the cells. We attach an annotation to a cell using an attribute property (d2s:attribute a qb:AttributeProperty). The text in the body of an annotation can provide an updated number for the cell (d2s:correctedPopulationSize a qb:MeasureProperty). Each annotation has an author (prov:wasGeneratedBy).

Second, the structure layer consists of an interpretation of table layout. Tables have common sections (see legend in Figure 2). Data cells contain the census counts. Headers and RowHeaders contain hierarchical entities that describe the numbers in their corresponding data cells. Properties do the link between data and RowHeaders, and

they are used as predicate names when linking a data cell with the RowHeader associated to it.

9 Although the UK census case (<http://cdu.mimas.ac.uk/index.htm>) is very similar to ours, including harmonization and Linked Data principles, no data model, tool or guideline seems to be available yet. 10 See <https://github.com/Data2Semantics/TabLinker>.

11 See <http://www.w3.org/TR/vocab-data-cube/>. 12 See <http://www.openannotation.org/spec/core/>.

13 See <http://www.w3.org/TR/prov-dm/>. :I skos:broader :I/E skos:broader :Nummer_{der}eroepsklasse :

Letter_{onderdeel}eroepsklasse_{d2s:populationSize};xd2s:dimensiond2s:dimension"1"8sd:int:14—151875—1874:M:O

To interpret table layout TabLinker needs tables to be marked, applying preset Excel styles to the main areas of the sheet (see Figure 2; different colors correspond to different styles). Headers and RowHeaders are read top-down and left-right, respectively. E.g., if A is a Header cell in top of the table subdivided in B, this is interpreted as A skos:broader B. Then, data cells are linked to their corresponding Header and Row-Header cells as Data Cube dimensions (CellN28 d2s:dimension A, B). One named graph is built per table.

Third, the harmonization layer lies on top and contains all mappings and classes necessary for internal linking (see Section 4).

7.3.2 Evaluation: querying the census

After successfully producing an RDF version of the census workbooks, RDF data was stored in a triplestore. The next step was to validate the RDF data, proving that meaningful queries for social historians can be issued against the RDF version of the census. In order to have a measure of success, we considered previous studies on the Dutch censuses [1]. In these studies contents of the census are analyzed, aggregated and plotted in order to explain social historical phenomena. Thus, showing that a number of SPARQL queries can produce the same results as the studies can prove the faithfulness of the RDF representation of the census, as well as the correctness of the query. This was very important to our experts. 14 We produced a number of geo charts representing results of two different queries that retrieve the same data on two different years. For several reasons (see Section 4) the same result cannot be achieved using one single query. 14 See <http://cedar-project.nl/visualizing-sparql-query-results-on-the-census/>.

:BENAMING_{van}de_{onderdeelen}der_{onderscheidene}eroepsklassen_{met}de_{aartoe}ehoorende_{eroepen}:Positie_{in}het_{eroep}

7.4 Requirements for Linked Humanities Data

Following a typical implementation of Linked Data principles in the Dutch historical census dataset did not produce ready-to-publish RDF data. Some of our requirements were not successfully met. Our take is that there is a lack of Semantic Web tools to tackle some characteristics of the Dutch census dataset found in other Humanities datasets as well. This section describes a set of requirements for such tools.

7.4.1 A world of mappings

In Section 2.1 we have presented some census variables (like occupations and geographical coverage) that do not completely match in labeling or meaning. The standard procedure to solve these non-matching entities, so that all data can be uniformly queried, is to use direct mappings. Harmonization can be seen as an additional metadata layer mapping mismatching labels, e.g. stating `A owl:sameAs B` or `A skos:exactMatch B` if A and B equal or exactly match, respectively. This solution, though, arises many other problems that make it impracticable in our case.

Size and complexity TabLinker produced 2,288 highly disconnected named graphs from the same number of tables, because these tables are self-meaningful and non-dependant between them. Thus, TabLinker did not produce a single triple connecting them. Any mapping is an interpretation to be placed in the harmonization layer.

A possible solution to this disconnection is to produce explicit mappings between related entities. Listing 1.1 shows how to do this with SPARQL queries on the census dataset: lines between 8 and 13 contain the graph pattern to retrieve men population per municipality. Assuming that this graph pattern remains the same across the tables, there are 2,288 different queries to refer to one variable with all its possible labels. If T is the set of tables and V the set of variables encoded, there are $T \times V$ ways for referring to the variables. In the worse case, all variables and tables are targeted to be queried at once, which for the census means $2,288 \times 122 = 279,136$ mappings. The cost of connecting the graphs via SPARQL is clearly unaffordable: too many queries derive in complex queries and expensive joins.

A possible solution to manage this number of mappings is to materialize them in the triplestore. For example, if the source layer contains the labels `RoomsKatholiekeKerk`, `RomsKatholic` and `VaticanChristelijk` for the variable `Religie` (religion), all three can be harmonized creating a new resource `Catholic` and issuing three triples with `Catholic` as subject, `skos:altLabel` as predicate and the labels as objects. This way, multiple interpretations become multiple graphs, and SPARQL can refer to these graphs instead of explicitly mapping each possible literal. However, this has two drawbacks: for a per-cell access,

```
PREFIX d2s: <http://www.data2semantics.org/core/> PREFIX d2sdata: <http://www.data2semantics.org/
data/VT187910_H1_marked/Noordbrabant/ > PREFIX ns1: <http://www.data2semantics.org/core/Noordb
PREFIX skos: <http://www.w3.org/2004/02/skos/core> SELECT ?place ?size WHERE ?cell d2s: isObs
```

Listing 1.1: SPARQL query for population of men per municipality, province of Noord-Brabant, year 1879.

queries still need to bind table dimensions to SPARQL variables (which produces expensive joins); and scalability is very complex, because domain experts have to link manually new literals with harmonizing classes.

One way of tackling with these interpretation graphs is using linksets. A linkset is a group of links that map two entities. Several linksets can be defined to describe different groups of mappings. For example, if author A1 provides the mappings `X skos:exactmatch Y`; `owl:sameAs Y` and author A2 does so with `X skos:related Y`;

skos:exactMatch Z, a linkset engine can solve the mapping (i.e. replace all references to X as references to Y or Z) if the user chooses criteria from A1 or A2.

As we have seen, Humanities RDF datasets are constantly subject to interpretation, context and non-destructive updates at fine-grained level. These can result in an arbitrary increase of the graph size and complexity, making graphs unmanageable for query engines and reasoners.

Inconsistency Humanities datasets have specific requirements regarding reasoning. We have already introduced the problem of graph size when adding context, which certainly affects reasoning.

But mappings and classes in the harmonization layer do not need to be consistent at all. Author A1, via annotations, can state `X owl:sameAs Y` and `Y owl:sameAs Z`, while author A2 can state `X owl:differentFrom Z` at the same time. Author A3 can define two disjoint classes U, V, while author A4 can create a common subclass of both. Though being contradictory, these harmonization mappings are true: this is how authors interpret the dataset. Despite these inconsistencies, reasoning facilities are still needed for inferred knowledge discovery. One possible solution is to contextualize inferencing, adapting the scope of the reasoner to make it consider only a subset of non-contradictory interpretations.

Concept similarity The described massive direct mapping scenario suggests that a more generic approach on concept similarity is necessary to improve management of concepts in the census and other Humanities datasets. We consider a theory of Concept Drift [14] as our definition for the meaning of a concept, its label, its intension and its extension.

Identity resolution [3] [6] plays a fundamental role as an harmonization operation in a Linked Data context. Very commonly, harmonizing values of a variable means grouping those that mean the same: different resources that have identical meaning have to be identified. In terms of Concept Drift theory, it is said that these concepts have identical rigid intensions (their fundamental properties are the same), while concept labels are different. Other scales of identity, like synonyms and homonyms, also exist here.

Many harmonization concepts are based on other concepts. Harmonization is also about taking two existing resources A and B and defining a new concept C in terms of A and B. For instance, a new concept Christian is introduced as a more abstract class holding both Catholic and Protestant as subclasses. But sometimes the hierarchy is specialized instead of generalized, or C only takes part of the extension of A and B (e.g. if A and B overlap and C takes the intersection).

Another case for harmonization occurs when labels remain the same over time. For instance, CivilWorker is used to refer to civil workers in 1879 as well as in 1920, but they are certainly not the same concept. While the labels are still the same, the meaning of the concept at its internal definition (its intension) has completely changed. A measure representing the intension shift of concepts is very valuable in this context [15].

Some concepts are destroyed or created, especially while trying to harmonize distant datasets in time. For instance, comparing the concept ComputerScientist in 1971 and 1791 makes no sense; likewise, PublicSpeaker does appear in a 1889 dataset, while it would rarely do in a 2012 one.

7.4.2 Beyond mappings

As shown, increase of size and complexity of census graphs due to direct mappings makes this strategy unsuitable for our case study: scalability of any system managing this knowledge base is severely compromised. This section exposes unmet requirements of the census (and other Humanities disciplines) to be considered as new challenges for Semantic Web technologies.

Dynamic ontologies Static, deterministic, agreed and formal denitions of meaning of concepts (the kind of denitions in which Semantic Web tools feel comfortable) are scarce in Humanities datasets. Formalization facilities provided e.g. by OWL do not at all with time-changing, non-deterministic, non-agreed and hardly formalizable concepts commonly found in the Humanities [5].

Despite Ontology Versioning [4,7], current tools lack time-awareness, i.e. the ability of the formalism to define differently a class of objects depending on the given moment in time, even if for that given moment no static denition has been explicitly attached. Many problems we encounter need variable, dynamic formalizations of concepts that change depending on a selected time frame. Having such denitions in the census case, the problem of having multiple, time-drifted labels for referring to a single occupation would no longer require direct mappings (neither big nor complex graphs), and concept similarity would be much easier to tackle. Interpretation, often derived from contested concepts and hermeneutics [16], provides subjective denitions when there is no consensus about the meaning of a concept. Authors are free to define concepts as they consider, usually in an inconsistent or contradictory fashion. Agreement can not be imposed, and all denitions shall coexist in a system that takes into account this definition multiplicity. A broader scope on integrating interpretation to current Semantic Web formalisms can be also a solution for ontology inconsistency in the context of the Humanities.

Partitions and counting One peculiarity of the census tables is that they only contain aggregations, i.e. counts of population meeting certain conditions. Thus, a census table can be seen as a partition of the total population, assuming that variables are not overlapping (i.e. every person is counted once and only once). Given the case of merging two variables of different tables that only match partially, how their numbers (i.e. data cells with the population counts) should be added? To what extent is possible to infer an individual (instance) of a variable (class) considering only data available in the table?

Format round-tripping In our case, census data has been converted from Excel workbooks into RDF named graphs. Many other formats are of interest of several user groups, like relational databases, CSV files or XML documents. The choice of RDF as an intermediate representation also has to meet the requirement of round-tripping: it must be possible to reconstruct the original Excel workbooks from the generated named graphs. Moreover, regardless of the transformations done between formats and the expressivity of their languages, internal equivalence with the original data has to be kept so that it can be reconstructed.

7.5 Conclusions

In this paper we have presented a conversion of the Dutch historical census dataset into RDF, taking into account key requirements from social scientists and historians, like guaranteeing a permanent access to original data and providing a framework for implementing harmonization. We have shown that harmonization is necessary in order to align the highly disconnected named graphs that TabLinker produces. Once done, visualizations could be greatly improved, consistently built and easily generated with appropriate uniform queries.

We have shown that the extraordinary size and complexity of these graphs explodes when direct mappings between disconnected (but related) entities are made explicit. Even proposing a three-tier architecture for organizing this knowledge into source, structure and harmonization layers, SPARQL queries that grab homogeneous data across multiple census tables turn extraordinarily complex. A standard Linked Data approach on the census also led us to inconsistency (due to contradictory harmonization statements) and to explore concept similarity.

Non-met requirements in the Dutch historical census case are our starting point to throw new challenges for Semantic Web technologies that could enormously help the Humanities solving their specific concerns on Linked Data. Awareness of time, interpretation and context as fundamental factors that greatly influence the definition of meaning of contested concepts can help creating a new version of the Semantic Web also suitable for Humanities data.

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Part III

Change of Meaning Over Time

Chapter 8

Semantic Similarity and Correlation of Linked Statistical Data Analysis

8.1 Introduction

There was this American who was afraid of a heart attack and he found out that the Japanese ate very little fat and almost did not drink wine but they had much less heart attacks than the Americans. But on the other hand he also found out that the French eat as much fat as the Americans and they drink much more wine but they also have less heart attacks. So he concluded that what kills you is speaking English [1]. While computers can assist us in discovering strong correlations in large amounts of statistical datasets, whether by chance or through sophisticated methods, humans (or sometimes also known as domain experts) still need to be critical about the results and interpret them appropriately. This implies that we are still very much involved in the process in discovering meaningful correlations by filtering through everything that is presented to us.

If we could however improve the situation slightly by having machines present us with only useful correlations from a random mass of correlations, then we can give more of our attention to what is interesting. Hence, our goal is to set a path towards identifying why some variables have a semantic link between them. Before we establish that, our ongoing approach (as outlined in this research and afterwards) will be to refute or cancel out things which may be in disguise for semantic similarity.

Therefore, we set our investigation with a workflow to experiment with Linked Statistical Datasets in the 270a Cloud [2]. We have first set our hypothesis to uncover the possibility that semantically similar variables or datasets need to incorporate semantically rich information in order to find thought-provoking correlations. Then, the question is, what do exceptional or intriguing linkages for semantic similarity look

like? We start with our null hypothesis by checking to see whether the dataset titles in World Bank indicators can help indicate strong correlations. Our results show that dataset titles by themselves or within a particular topic area is not a good indicator to predict correlations.

8.2 Methodology

We first state our research design and hypothesis, then discuss how we employed Linked Statistical Data (LSD) and Semantic Similarity approaches for a workflow in our LSD Sense [3] implementation.

8.2.1 Research design

Research problem: Why do machines have difficulty in revealing meaningful correlations or establishing non-coincidental connection between variables in statistical datasets? Put another way: How can machines uncover interesting correlations?

Over this ongoing investigation, we want to uncover some of the fundamental components for measuring and declaring semantic similarity between datasets, in order to better predict relevant strong relationships. Can semantic relatedness between datasets imply statistical correlation of the related data points in the datasets?

8.2.2 Hypothesis

Given our research question, we would like to propose a viable research hypothesis, followed by our investigation with the null hypothesis:

H: If the absence of semantically rich connection between datasets is inadequate to distinguish meaningful relationships, then making relevant information about dataset connectivity will improve predicting dataset correlations by observing their semantic similarity.

H: There exists a significant relationship between the semantic similarity of statistical dataset titles and the correlation among those datasets, because dataset titles can indicate rich connectivity.

We set the significance level to 5

8.2.3 Linked Statistical Data and Semantic Similarity

The RDF Data Cube vocabulary does not only allow one to express statistical data in a Web exchangeable format, but also to represent the (semantic) links within those statistical data. This ability poses some new interesting research questions around the relationship between the statistical and semantic relatedness of datasets. We are interested in the interplay of statistical correlation of LSD and their semantic similarity,

in order to answer questions like: Does correlation between statistical datasets imply some kind of semantic relation? Do certain semantic links imply the existence of correlation? We propose a generic workflow for studying whether or not this relation between correlation and similarity holds for arbitrary LSD. We aim at generic correlation and similarity measures, and our workflow enables the use of any correlation and similarity indicators. For the specific goal of this paper, though, we stick to the use of Kendall's correlation coefficient and Latent Semantic Analysis (LSA) similarity.

8.2.4 Workflow

Based on preliminary experimentation from data acquisition to analysis, we have created the LSD Sense workflow:

Create hypothesis

Determine datasets and configurations

Get metadata of datasets.

Get each dataset's observations.

Create correlations and other analysis for each dataset pair combination.

Create dataset metadata subset for semantic similarity.

Create semantic similarity for each dataset pair combination.

Create correlation and other analysis using variables semantic similarity and correlation of LSD.

Test and verify hypothesis.

Analysis.

8.2.5 Implementation

We have an implementation of the LSD Sense workflow which can be used to both, reproduce our experiments, as well as run it on new input datasets. With the exception of determining which datasets to inspect, and the system configuration, LSD Sense is automated.

Semantic Correlation

The semantic similarity algorithm is based on a Latent Semantic Index (LSI) [4]. We use the dataset titles to check for their similarity. Essentially, LSI puts each dataset title into a cluster. The number of clusters can be adjusted (default to 200). It remains as an open research question as to what it should be. Generally, research has demonstrated that optimal values depend on the size and nature of the dataset [5]. We use gensim [6] in our Semantic Correlation [7] implementation for LSD Sense.

Concerning the quality of the dataset titles, it is possible to come across datasets that differ only by one word e.g., male, female. This potentially lowers the accuracy to

differentiate datasets. As mentioned earlier, we removed the attribute information from the dataset titles with the assumption that it reduced noise.

8.3 Experiment

Two experiments were conducted using the same workflow. Experiments differed only by their input data. In the first experiment, the analysis was done for a particular reference year over all available datasets. In the second experiment, however, we restricted the data further for only a particular dataset domain (topic), thereby making it possible to compare whether a control over a topic can be significant for semantic similarity of the dataset titles.

8.3.1 Data

We decided to conduct our experiment on a simple dataset structure, containing two dimensions; reference area, and reference period, and one measure value for its observations, where the World Bank indicators was a good candidate from the 270a Cloud. The rationale for using only one dataspace (at this time) was to remain within a consistent classification space to measure semantic similarity. We fixed the reference period to 2012, and datasets that are part of World Bank's education topic. We have identified one downside concerning the data quality i.e., the attribute/unit information was incorporated as part of the dataset title, usually as a suffix within brackets. We dealt with this by removing the attribute information from the titles as part of preprocessing in the semantic similarity phase.

8.3.2 World Bank Indicators workflow

The workflow of our experiment is summarized as follows:

Correlations for each dataset pair

We retrieved the 2012 World Bank Indicators datasets, 3267 in total, via SPARQL queries from the World Bank Linked Dataspace [8]. The correlations were computed using R, the statistical software, by joining each dataset pair by their reference area (one of the dimensions of the dataset structure), and using their measure values for the correlation coefficient. Based on preliminary inspection for normality distribution on sample datasets, we noted that observations did not come from a bivariate normal distribution. Hence, we computed Kendall's rank correlation coefficient in our analysis. Initially we computed and stored the correlations for dataset pairs with a sample size, $n > 10$, resulting in 2126912 correlation values. The information on the analysis we generated consisted of the following headers: datasetX, datasetY, correlation, pValue, n, where datasetX and datasetY are the identifiers for each dataset pair that is

being compared. We later filtered sample values, $n < 50$, for our threshold for significance. The population size i.e., the number of potential reference areas that can have an observation, is 260. That is the number of reference area codes in the World Bank classification, however, it is not known as to which reference areas may occur in a given dataset beforehand. We retained majority of the computations in any case, giving us the possibility to do better pruning in the future, in light of more information.

Semantic similarity for each dataset pair

Before doing the semantic similarity, we first took an unique list of the dataset identifiers from datasetX and datasetY so that what is to be checked for their similarity is only in relation to those datasets, as opposed to the complete set of datasets which we originally retrieved. At this point, we have 2200 unique datasets. The similarity was measured based on dataset titles. They are in short sentences e.g., Mortality rate, infant (per 1,000 live births). After minor preprocessing e.g., removal of the text pertaining the unit within brackets, it was left with Mortality rate, infant. The semantic similarity algorithm is based on LSA. Essentially LSA puts each dataset title into a cluster (default number is 200). The resulting headers of the output was: datasetX, datasetY, similarity.

Correlation analysis with variables semantic similarity and correlation of dataset

We then took the absolute values for both variables; |similarity|, |correlation| (caring only for the strength of the relationships as opposed to their directionality). We then filtered both similarity and correlation values < 0.05 and > 0.95 , as well as correlation values with $p\text{-value} > 0.05$, for reasons to exclude potential outliers, or misleading perfect relations, as well as to exclude insignificant correlations. The final correlation and scatter plot was generated by joining the similarity and correlation tables on datasetX and datasetY columns. Finally the correlation of the final data table was conducted using the Kendall method as the data had a non-normal distribution and we were not interested in modeling (line fitting).

The second experiment followed the same procedure for the analysis, but considering only the datasets associated with the topic education for the same reference period.

8.4 Results

All of the experiment results are available at the LSD Sense GitHub repository, and can be reproduced. Table [Experiment results] provides our findings, with Figures [1] and [2]:

Experiment Results All topics One topic (education) Datasets are from 2012 World Bank indicators. n is the number of dataset pairs with semantic similarity and correlation as variables. Correlation 0.182 0.227 p-value $< 2.2e-16$ $< 2.2e-16$ n 92819 33184

Figure of scatter plot showing 2012 World Bank indicators with all topics Figure 1:
2012 World Bank indicators with all topics

Figure of scatter plot showing 2012 World Bank indicators with topic education Figure 2:
2012 World Bank indicators with topic education.

Given that both experiments resulted in p-values that are statistically significant and that the strength of the correlation values are weak, we reject our null hypothesis. For extra measure, we can also verify the meaninglessness by looking at the plots. There is nothing interesting to see here. We will move along with our alternative hypothesis.

8.5 Related Work

Linked Statistical Data Analysis [9], explores a way to reuse statistical linked dataspaces, federated queries, and generation of statistical analyses e.g., regression, for humans and machines. The stats.270a.info [10] service stores computed analysis, and makes it possible for future discovery.

Towards Next Generation Health Data Exploration: A Data Cube-based Investigation into Population Statistics for Tobacco [11], presents the qb.js [12] tool to explore data that is expressed as RDF Data Cubes. It is designed to formulate and explore hypotheses. Under the hood, it makes a SPARQL query to an endpoint which contains the data that it analyzes.

Generating Possible Interpretations for Statistics from Linked Open Data [13] talks about the Explain-a-LOD tool which focuses on generating hypotheses that explain statistics. It has a configuration to compare two variables, and then provides possible interpretations of the correlation analysis for users to review.

Using Linked Data to Evaluate the Impact of Research and Development in Europe: A Structural Equation Model [14], presents the feasibility of combining different LOD sources to assess the impact of one variable over others.

Spurious Correlations [15] reveals correlations that are not genuine for practical use. In other words, the correlations are type I errors. It emphasizes on the importance for humans to be critical of random correlations, and to investigate whether there is a direct relation between the variables.

Ontology Matching [16] is perhaps the most mature field in the Semantic Web dealing with the general problem of finding semantically related entities of ontologies and Linked Data, although resources like WordNet and DBpedia are also related.

These studies and engineering efforts have created, inspected, and hypothesized possible correlations. However, the missing gap in research is that there is no integrated study on how semantic relatedness between datasets may enhance the detection of meaningful or useful correlations in statistical data. Our contribution is the investigation of highly probable elements which would lead to better prediction of interesting correlations by employing linked statistical datasets and semantic analysis.

8.6 Conclusions and Future Work

We believe that the presented work here and the prior Linked Statistical Data Analysis effort contributes towards strengthening the relationship between Semantic Web and statistical research. What we have set out to investigate was to minimize human involvement for discovering useful correlations in statistical data. We have implemented a workflow in which we can automate the analysis process, from data retrieval to outputting analysis results for candidate semantic linkages in Linked Statistical Data.

We have evaluated our results by testing and verifying the null hypothesis which we have put forward. While it turned out that the semantic similarity between datasets titles were not useful to determine strong and meaningful correlations — which is a useful finding, in any case — it left us with the remaining alternative hypothesis that can be used in future research.

Possibly fruitful future work might want to run a similar experiment with the semantic similarity of dataset descriptions, test manually configured useful relations for a controlled set of datasets, or looking into interlinked topic domains across linked dataspaces.

Where is interestingness hidden?

8.7 References

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Chapter 9

Radical Transitions of Concept Extensions

9.1 Introduction

Availability of Linked Data on the Web has notably increased over the last decade^{1,2}. Openness and reachability have encouraged developers to build web applications and mashups retrieving, comparing and composing these data. *Concepts* are the basic units of thought and the central entities of datasets published as Linked Data. Although they are often assumed to be stable, concepts can change their meaning over time, space, and other dimensions. We call this phenomenon *concept drift*, and it can have a severe impact on results of applications that combine different data sources. We are interested in the analysis of concept drift and how important changes in data occur over several dimensions. In this paper we propose a formal framework for the study of concept drift.

Concept drift refers to the change of meaning of concepts, as the reality they represent changes continuously. Concept drift can have drastic consequences on the use of a concept in an application. A concept may replace the meaning of other concepts, or other concepts can take over its meaning. This produces errors in data retrieval that are very difficult to trace and address. Concept drift can happen at the concept identifier level (*label drift*), in the basic properties of the concept (*intensional drift*), or to the things the concept refers to (*extensional drift*) [166].

We are especially interested in concept drifts that National Statistical Offices (NSOs) are concerned about. Suppose a simple aggregated dataset like shown in Table 9.1 with the concepts *municipality*, *occupation* and *population* describing demographic data. These and larger tables exist for a great amount of territories, time frames and socio-economic variables. Although some of these tables are ordered in a way that facilitates

¹<http://lod-cloud.net/state/>

²<http://eurostat.linked-statistics.org/>

the analysis of concept drifts (e.g. time series are suitable for concept drift over time), some of them lack a meaningful ordering supporting their analysis. Detecting and tracing the existence of important changes in such data over several dimensions, including time, space and economic factors, is a complex problem. Generic solutions would enable large scale discovery of changing concepts and, consequently, would improve the coherence of analyses in these datasets.

The problem has been only partially solved and scarcely formalized. Other approaches propose solutions (see Section 10.2), but these rely on *a)* availability of data about individuals, or *b)* some fine-grained formalization of the concepts and their intensions and extensions. In our datasets some or none of these may be available: they consist of aggregated data, usually counts of observations, for a given concept of interest.

In this paper we extend our previous work on detecting extensional concept drift using statistical tools [?, ?]. The main contribution of this paper is a generic framework that addresses the problem of detecting meaningful changes in concept extensions over different dimensions. In this framework we define tables, rewrites, selections and orderings to manipulate datasets:

- Containing *aggregated* data, i.e. sums or counts that cannot be resolved to the original individual observations;
- Modelled with concepts used over different dimensions, like *time*, *location* or *economic factors*, with some given notion of identity (i.e. provided by experts);
- Where such dimensions are ordered, and orderings are possibly related to semantics;
- Where the different orderings of concepts and tables affect the meaningfulness of changes.

Under these assumptions, we propose the notion of *radical transitions* to formalize meaningful changes in concept extensions. We identify a basic condition for radical transitions to be semantically meaningful, which we call the *smoothness assumption*.

We apply our framework to two different case-studies on statistical datasets. The first consists of historical Dutch census data, where we leverage the temporal dimension to study concept drift over time. The second focuses on current Australian and French censuses, where no temporal variants of the same concepts exist, and only economic dimensions are available. We investigate the usefulness of our generic framework to discover meaningful changes in concepts over these multiple dimensions. We study the validity of our results, and how it depends on the smoothness assumption of the chosen dimension. Interestingly, the radical transitions we detect have coherent socio-historical and economical causes.

This paper is organised as follows. In Section 10.2 we describe the state of the art in developed methods for concept drift. In Section 9.3 we provide some background on the concept drift approaches and the statistical tools we use. In Section 9.4 we define our framework for the study of radical transitions. In Section 9.5 we describe the application of radical transitions in two different case-studies, providing evaluation and results. Finally, in Section 10.7 we establish some conclusions and further work.

		Population			
		21-30 years		31-40 years	
		Male	Female	Male	Female
2*Amsterdam workers	Shoemakers	217	121	288	209
	Bakers	82	31	122	46
2*Leiden workers	Shoemakers	21	2	34	43
	Bakers	9	18	8	75

Table 9.1: A table T with concepts municipality, occupation, population.

9.2 Related Work

Concept drift is a very active research topic in Machine Learning [154], where it is defined as the situation in which the statistical properties of a target variable change over time in unforeseen ways. Learning from data streams in such a situation requires a concept drift detection method [72, 90, 95, 133]. On the Semantic Web, concept drift relates to the study of the dynamics of meaning. In ontologies, this is addressed in ontology change and evolution management ([?, 102, 116]). Fanizzi et al. [92] propose a method based on clustering similar instances. But Description Logics have also addressed the related problem of detecting differences between ontologies [99]. To the best of our knowledge, the work of Wang et al. [166, 124] is the only concept drift formalization in a Semantic Web setting. Meroño-Peñuela et al. [?, ?] describe the basic statistical tools and methods in which we base our work.

9.3 Background

9.3.1 Concept Drift

The world is continuously changing, and concepts also change over time. A concept refers to different objects, real or abstract, at different points in time. For instance, the concept `Manager` refers to different types of occupations in 1795 and in 2013. We use the formalization framework described by Wang et al. [166] in order to address change of meaning over time.

The *meaning of a concept* C is a triple $(\text{label}(C), \text{int}(C), \text{ext}(C))$, where $\text{label}(C)$ is a string, $\text{int}(C)$ a set of properties (the *intension* of C), and $\text{ext}(C)$ a subset of the universe (the *extension* of C).

All the elements of the meaning of a concept can change. If two variants of a concept at two different times have the same meaning, there is no concept drift. We define intensional, extensional, and label similarity functions $\text{sim}_{\text{int}}, \text{sim}_{\text{ext}}, \text{sim}_{\text{label}}$ to quantify meaning similarity. Each of these functions has range $[0, 1]$, and a similarity value of 1 indicates equality.

A concept has extensionally drifted in two of its variants C' and C'' iff $\text{sim}_{\text{ext}}(C', C'') \neq 1$. Intensional and label drift are defined similarly.

To apply this framework of concept drift it is required to define intension, extension and labelling functions, and to define similarity functions over intension, extension and labels.

9.3.2 Statistics

The Wilcoxon signed-rank test [168] is a statistical hypothesis test used when comparing two related samples, matched samples or related measures to assess whether their population mean ranks differ. It is used instead of other statistical tests, such as Student’s t-test, when the population cannot be assumed to be normally distributed.

It works under the following assumptions:

- Data are paired and come from the same population
- Each pair is chosen randomly and independently
- Data are measured, at least, on an ordinal scale

More practically, the Wilcoxon test is used to test at some significance level s (typically $s = .05$) if two given samples have identical data distributions. To run this test, the null hypothesis H_0 is that the two given samples are, indeed, identical populations. Whenever the test returns a p-value lower than the significance level s , we can reject H_0 , meaning that the two compared samples are non-identical.

9.4 Radical Transitions

In this section we propose a theoretical framework that combines the definition of extensional concept drift and the statistical test described in Section 9.3. Its purpose is to detect extensionally drifted concepts over generic dimensions through *radical transitions*. It is suitable for tabular data consisting of numerical aggregations. First, we define *tables* as the formal structures where concept drift detection is to be applied, and their relationships with *concepts* and concept extensions. Second, we define what an *ordering* of these tables is, according to some *dimension*. Finally, we distinguish, from all dimensions, those satisfying the *smoothness assumption*, as a necessary condition to detect *radical transitions*. Intuitively, radical transitions are meaningful changes in concept extensions that indicate drift of meaning.

9.4.1 Tables, Rewrites, Selections and Orderings

The datasets we consider come in tables with different format, structure and layout (e.g. hierarchical columns and rows, see Table 9.1). To simplify notation we introduce a table canonical form, a rewrite function and the notion of table selection.

Assume a dataset organized in tables of the canonical form $T = \{(C, ext(C))\}$, where C is a concept and $ext(C)$ is the extension of C given as a vector of numerical values $(e_1, \dots, e_n) \in \mathbb{R}^n$.

A table T is in *canonical form* iff $T = \{(C, ext(C))\}$ where C is a concept and $ext(C) = (e_1, \dots, e_n) \in \mathbb{R}^n$.

Intuitively, these values represent observations that characterize the concept C . For example, $T = \{(\text{males}, (1, 5, 2, 3, 4)), (\text{females}, (2, 4, 8, 1, 7))\}$ is a valid table.

If source data come in a non-canonical tabular form $T_c \neq T$, we define *table rewrites* such that T_c is represented as another table T in canonical form.

The function $f_{rw} : T_c \mapsto T$ is a *table rewrite* iff f_{rw} is bijective and T_c, T are isomorphic. We also say that f_{rw} is *lossless*.

In some cases, we may be interested in a subset of T instead of the whole table.

The function $f_s : T_1 \mapsto T_2$ is a *table selection* iff $C_2 \subseteq C_1$ and $ext(C_2) \subseteq ext(C_1)$ for $T_1 = (C_1, ext(C_1))$ and $T_2 = (C_2, ext(C_2))$. Table selections are also tables. We also write $T_2 \subseteq T_1$.

Given two tables $T_1 = (C_1, ext(C_1)), T_2 = (C_2, ext(C_2))$ with, we introduce the following table order relation:

$$T_1 \leq T_2 \iff d_x(C_1) \leq d_x(C_2) \quad (9.1)$$

where $d_x : C \rightarrow \mathbb{R}$ is a *dimension function* that maps a concept C with its corresponding dimension value for a given dimension x . d_x must be thus defined for all $C \in T$.

Typical examples of such dimensions are *time*, *spatial location* or *population density*. Each of these dimensions define specific dimension functions d_t, d_l, d_{pd} that provide orderings for the given tables, applying Equation 9.1. d_t sorts the tables in ascendant order according to time (i.e. produces a time series), d_l sorts them according to geographic locations, and d_{pd} sorts them according to population densities of places contained in the tables.

9.4.2 Smoothness Assumption

Intuitively, a dimension function fixes the basis in which the study of the variation of concept extensions is to be performed. Accordingly, a dimension function d_x and its corresponding ordering can be defined, in general, for any arbitrary dimension x that applies to a concept C . The best studied and most intuitive of these dimensions is *time*, but we will discuss other orderings later in the paper.

However, not all dimensions are suitable for an effective extension concept analysis. We argue that, in order to perform meaningful analyses of change in concept extensions, dimensions have to meet criteria we call the *smoothness assumption*.

Given a concept $C \in T$ and $\epsilon, \delta > 0$, the dimension function $d_x(C)$ meets the *smoothness assumption* iff $|d_x(C') - d_x(C'')| < \delta \Rightarrow 1 - sim_{ext}(C', C'') < \epsilon$.

The smoothness assumption depends on the concept C . Hence, *time* and *population density* are good dimension candidates for concepts which extensions are expected to suffer minor variations over time and density, such as the socioeconomic concepts of

occupations or wealth. On the other hand, *spatial locations* do not meet smoothness for such concepts, since variations of extensions over these might vary more than ϵ for extremely different economical profiles.

9.4.3 Radical Transitions

We have already defined the extension of a concept $ext(C)$ as a set of numerical observations that describe C . We also interpret similarity of extensions in a statistical setting. Given a concept C and two of its variants, C', C'' , we define the extension similarity function $sim_{ext}(C', C'')$ as follows:

$$sim_{ext}(C', C'') = \text{wilcox.test}(ext(C'), ext(C'')) \quad (9.2)$$

The extension similarity function $sim_{ext}(C', C'')$ is defined as the function that returns the probability that the two variants C' and C'' of C have identical populations. This function, as discussed previously in Section 9.3.2, is the Wilcoxon signed-rank test. Hence, we assume that the extension of C has drifted between C' and C'' iff the populations of C' and C'' are non identical, that is, whenever $sim_{ext}(C', C'') \neq 1$ (see Section 10.3.1). In such case, we conclude that there is extensional concept drift between C' and C'' . Since the Wilcoxon test is symmetric, we assume $sim_{ext}(C', C'') = sim_{ext}(C'', C')$.

Given an arbitrary number n of tables $\{T_n\}$ and a dimension function d_x , it is possible to order these tables (see Section 9.4.1) and evaluate the extension similarity function $sim_{ext}(C_i, C_j)$ as defined in Equation 9.2 over all concepts of each ordered pair of tables:

$$\begin{aligned} &sim_{ext}(ext(C_i), ext(C_j)), \\ T_{i,j} = (C_{i,j}, ext(C_{i,j})), 1 \leq i \leq j \leq n \end{aligned} \quad (9.3)$$

This evaluation returns whether there exists extensional concept drift for the two concept variants C_i, C_j of T_i, T_j , according with the Wilcoxon test definition (see Section 9.3.2). When extended to all $\{T_{i,j}\}$ ordered pairs, the evolution on the change of concept extensions according to dimension x is traced. The interesting points of this evolution are the pairs C_i, C_j such that $sim_{ext}(C_i, C_j) < s$, where s is typically set to .05. We define such points as *radical transitions*.

Given a concept $C \in T$, two of its variants C', C'' and a constant $s \in [0, 1]$, a *radical transition* exists between C' and C'' iff $sim_{ext}(ext(C'), ext(C'')) < s$.

Different dimensions x allow analyses of the change of concept extensions over time, space, economic factors, etc. In Section 9.4.2 we formulated a smoothness condition of orderings of tables for specific concepts. This definition gives rise to our main hypothesis, namely that radical transitions of extensions are meaningful indicators for concept drift if the data ordering satisfies the smoothness assumption.

Given a concept $C \in T$ and a dimension function $d_x(C)$, a radical transition of two variants C', C'' of C is meaningful iff $d_x(C)$ satisfies the smoothness assumption.

9.5 Case-Studies

In this section we study the notion of *radical transitions* (see Section 9.4) to identify extensional concept drift in two different case-studies and three datasets published in the Semantic Web in RDF Data Cube [87] as Linked Data [?]. We use the statistical framework R to perform the experiments.

9.5.1 Concept Drift Over Time

In the first case-study we detect radical transitions of concept extensions in the Dutch historical censuses. This dataset comprises 507 Excel workbooks, containing 2,288 census tables. These tables have been created by hand, using the digitized images of the original census books. Census data refers here only to aggregated data (i.e. counts of people meeting certain conditions). Microdata (i.e. data about individuals) is not available in this dataset. Each census table describes a portion of the population, occupation or housing census of a certain year. In this experiment we use tables of the occupational census in the province of Noord-Holland in the years 1889 and 1899. Due to the late industrialization of the Netherlands in the 19th century, data of this time and region are inclined to contain more drastic changes in the occupational landscape. Tables have been converted to RDF Data Cube using TabLinker³ [?].

The dataset is rich in demographic and socioeconomic concepts. Table 9.2 shows a sample of data queried using SPARQL⁴ and loaded in R via the SPARQL R package [158]. We select the concepts $C = \{age\ range, position, gender, marital\ status, municipality, occupation, population\}$ for the two years.

All values have been normalized, some with the help of external classification systems (e.g. occupations through codes of the Historical International Standard Classification of Occupations, HISCO). These mappings are manually established by experts, pairing each occupation appearance in the census tables with one (and only one) HISCO code, providing a notion of identity between variants of concepts.⁵ HISCO codes follow a tree-like structure: code 12310, for instance, refers to occupational titles under the micro group 12310 (*Notary*), unit group 123 (*Notaries*), minor group 12 (*Jurists*), and major group 1 (*Professional, technical and related workers*). Only mappings with micro groups (i.e. five number codes), which are the leaves of the tree, are allowed.

Given following dimension function based on *time*,

$$d_t(C) = \begin{cases} 1889 & \text{if } C \in T_1 \\ 1899 & \text{if } C \in T_2 \end{cases} \quad (9.4)$$

the goal is to trace extensional concept drift through radical transitions of the concepts in *occupations* in the ordered table pair T_{1889}, T_{1899} . Hence, we define the ta-

³See <https://github.com/Data2Semantics/TabLinker/> and <https://github.com/CEDAR/DataDump/>

⁴SPARQL endpoint at <http://lod.cedar-project.nl:8080/sparql/cedar/>

⁵See <https://github.com/albertmeronyo/ConceptDrift/>

Age range	Position	Gender	M. status	Municipality	Occupation	Population
36-50 years old	A	Male	Married	Velsen	21240	3
51-60 years old	B	Female	Unmarried	Zaandam	44120	1
23-35 years old	C	Female	Unmarried	Haarlem	30000	2
36-50 years old	A	Male	Married	Weesp	95000	5

Table 9.2: Sample rows of the data frame for the first case-study (Dutch historical occupations over time). The first two belong to the 1889 data, last two to 1899. HISCO code 21240 represents *contractors*, 44120 are *manufacturer’s agents*, 30000 are *civil workers* and 95000 are *house finishers*.

Gender	Age range	Location	Labour status	Population
Male	15-19 years	New South Wales	Employed, Total	78743
Female	25-29 years	Victoria	Not in the labour force	40045
Men	25 to 29 years	Nord	Unemployed	8.87e+04
Women	65 years and over	Paris	Other inactive	1.10e+03

Table 9.3: Sample rows of the data frame for the second case-study, current Australian and French censuses respectively.

ble selections $T_{1889_{occ}}, T_{1899_{occ}}$ and for each occupation C we want to know whether $sim_{ext}(C', C'') < s$ for the variants $C' \in T_{1889}, C'' \in T_{1899}$ (see Definition 9.4.3). Intuitively, this traces to what extent the extension of occupational concepts changed in Noord-Holland during the period 1889-1899.

9.5.2 Demographic Unbalance

The second case-study consists in detecting radical transitions in concept extensions in the RDF Data Cube versions of the Australian census of 2011 and the French census of 2010.⁶ Table 9.3 shows a sample of the queried data using SPARQL⁷ and loaded in R via the SPARQL R package [158]. We select the concepts $C = \{gender, age\ range, location, labour\ status, population\}$.

In the first case-study (see Section 9.5.1) we define the dimension function based in *time*. In the second case-study we propose the ordering derived from choosing alternative dimension functions that also satisfy the smoothness assumption (see Section 9.4.2). We use the dimension functions *GDP per capita* of the Australian states to order the Australian tables, and *population density* of the French *departements* to order such occurrences.

$$d_{gdp}(C) = GDP(S), S \in \{\text{Australian states}\} \quad (9.5)$$

$$d_{pd}(C) = PD(D), D \in \{\text{French departements}\} \quad (9.6)$$

Such functions define the tables $\{T_i\}, i \in \{1, \dots, 8\}$ and $\{T_j\}, j \in \{1, \dots, 96\}$ for the Australian and French case, respectively, and the corresponding orderings according to

⁶Source code at <https://github.com/albertmeronyo/ConceptDrift/blob/master/stats/semstats-challenge.R>

⁷SPARQL endpoint at <http://lod.cedar-project.nl:8080/sparql/semstats/>

the defined dimension functions.

The goal in this second case-study is to study the extensional drift of the concept $C = \{\text{youth unemployment}\}$ in both censuses. Given the previous sets of ordered tables, for $C = \{\text{youth unemployment}\}$ we want to know whether $\text{sim}_{ext}(C_i, C_j) < s$ for the variants C_i, C_j of each ordered pair $T_i \leq T_j$ (see Definition 9.4.3). Intuitively, this traces to what extent the extension of youth unemployment varies among richer or more dense regions, respectively, of these countries.

9.5.3 Evaluation

Census data of the two case-studies come in non canonical form (see Table 9.1). In order to generate canonical tables from RDF Data Cube datasets, we perform some SPARQL queries that return CSV-like results, which are much closer to canonical tables defined in Section 9.4.1. We implement the table rewrite function f_{rw} through SPARQL templates for RDF Data Cube data [?].

We normalize the values of nominal variables through standardization, and we use external data sources to normalize municipalities and occupations. We also perform some cleaning, removing incomplete rows, partial total population aggregations, aggregations at the province level, and aggregations of smaller villages that only appear in one of the two tables.

Table selections (see Section 9.4.1) are used in both case-studies to filter the concept extensions of interest in each case, and are performed via R statements that select only a specific subset of concepts and concept extensions (see Listing 9.1).

Table orderings (see Section 9.4.1) are generated differently in the two cases. In the first case (Dutch censuses), we implement the function $d_t(C)$ to retrieve the corresponding year of a concept from external dataset metadata. In the second case (Australian/French censuses) we perform live queries to DBPedia⁸ to implement the dimension functions. The function $d_{gdp}(C)$ retrieves the *gross domestic product (GDP) per capita* of all states in the Australian case. In the French case, the function $d_{pd}(C)$ is implemented in the following way: we ask for the area a and total population p for each *département*, and then we derive the corresponding *population density* as $pd = \frac{p}{a}$. The results provided by the dimension functions are used to generate the corresponding table orderings in the Dutch, Australian and French cases.

Detection of radical transitions is performed via the built-in `wilcox.test` function of R. For example, the occupation concept *Contractors*, which has been normalized in the first case-study with the HISCO code 21240 (see Table 9.2), has the extensions shown in Listing 9.1 in tables T_{1899}, T_{1899} . We run the built-in `wilcox.test` function using these extensions as input, concluding that the population of contractors in Noord-Holland in the years 1889 and 1899 are statistically identical populations ($p > 0.05, N = 42$, Wilcoxon signed-rank test). Consequently, there is no radical transition in this case. We proceed analogously for the Australian and French cases.

⁸<http://dbpedia.org/sparql>


```

1 > x <- df1889[df1889.hisco$HISCO == '21240',
2   'population']
3   3  1  1  2  2  1  3  1  1 10 10  5  2  1
4   1  1  1  4  1  4  1  1  1  2  1  8  2  1
5   2  5  1 50  1  1  1  1  1  1  2  1  1  1
6 > y <- df1899[df1899.hisco$HISCO == '21240',
7   'population']
8   20  3  1  3  1 30 10  5  4  1  4 10  1  9
9   1  3  1  8  1  4  4  1  1  2  1  1  1  3
10  1  1  1  1  1  1  1  1  1  2  1  1  1  2
11 > wilcox.test(x, y)
12
13   Wilcoxon rank sum test with continuity
14             correction
15
16 data:  x and y
17 W = 830, p-value = 0.6063
18 alternative hypothesis: true location shift
19 is not equal to 0

```

Listing 9.1: Arrays of concept extensions in 1889 and 1899 in Noord-Holland of HISCO code 21240 (*contractors*). We apply the procedure of Listing 9.1 iteratively over all variants of the same occupational concept C in ordered tables T_{1889}, T_{1899} according to the dimension function d_t . For the Australian and French cases, we apply the same procedure iteratively between all pairs of ordered tables T_i, T_j according to the dimension functions d_{gd} and d_{pd} , respectively.

9.5.4 Results

We first analyse results of the first case-study, the Dutch historical censuses.

Tables T_{1889}, T_{1899} have 57 and 88 non-common HISCO codes, respectively. 42 out of the 217 (19.35%) common HISCO codes are found to have p-values under the $s = .05$ threshold, resulting in rejection of H_0 (see Section 9.3.2) and consequently denoting radical transitions.

Table 9.4 shows the results of radical transitions in extensions of occupational concepts. We find radical transitions in 42 out of 217 (19.35%) historical occupational titles of the Dutch censuses between the years 1889 and 1899.

These results are consistent with historical and socioeconomic analyses performed by expert historians. More concretely, the results show evidence of the slow industrialization of the Netherlands in the late 19th century. Entrepreneurs put emphasis on trade rather than industry, although results show important variations in both groups. Interestingly, occupations with more radical transitions have relationships with the usage of one or another *technology* or specific *machinery*, which is a known factor of change in the history of work. Not surprisingly, *loaders of ships* are the top occupation which radically changed the most. In line with this, the general work groups related to *production, transport and operators* also appear on the top of the extension change list.

The second case-study, the Australian and French non-temporal censuses, requires a different analysis because of the number of tables.

HISCO	Occupation	p-value				
97125	Loader of ship, truck, wagon or airplane	1.83e-10	Group 7, 8, 9 5 4 2	Type Production, trans- port, operators Service workers Sales workers Administrative and managerial	p-value 2.03e-19 1.88e-12 2.94e-08 4.20e-08	
21110	General manager	4.23e-09				
41025	Working proprietor (wholesale, retail trade)	1.52e-08				
79100	Tailor	7.75e-07				
57030	Barber, hairdresser	1.17e-04				
88010	Jeweller	1.84e-04				
(a) Occupations with stronger ext. drift.			(b) Major groups with stronger ext. drift.			

Figures 9.1a and 9.1c show the resulting p-values that indicate whether there is a radical transition of *youth unemployment* between the different Australian and French regions (p-value < s) or, on the contrary, the concept remains stable. The horizontal axis represents the tables T_i , and the vertical axis represents the assigned $sim_{ext}(ext(C_{i-1}, C_i))$ values. The points below the s horizontal line that correspond to table T_i denote radical transitions between T_{i-1} and T_i . Since the first ordered table has no previous one to be compared to, the first p-value is always $sim_{ext}(ext(C_1), ext(C_1)) = 1$.

However, these plots display only the local difference of radical transitions. In order to visualize an evolution of such transitions in a global scale, for each application k of Definition 9.4.3 (see Section 9.4.3) we compute the distance function (typically $s = .05$)

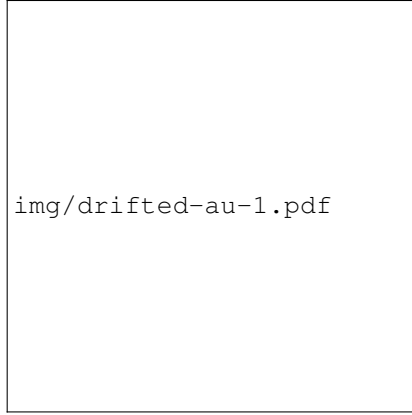
$$d_k = \begin{cases} p_{k-1} - \alpha(p_k) & \text{if } p_k < s \\ p_{k-1} & \text{if } p_k \geq s \end{cases}$$

where α is a function that magnifies the distances in case of drift. Results after the evaluation of this distance function are shown in Figures 9.1b and 9.1d. According to these figures, we find radical transitions in 5 out of 8 (62.50%) Australian regions and 2 out of 96 (2.08%) French *departements*.

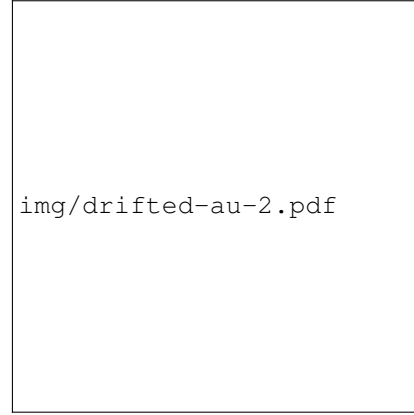
Figures 9.1b and 9.1d show meaningful results on the use of the dimensions *GDP per capita* and *population density* in the absence of *time* to track the evolution of radical transitions in youth unemployment. In the Australian case, the extensions of this concept tend to vary in the less rich regions. There is an interesting stability of concept extensions in the middle area, where the 3rd to 7th less rich regions do not show much differences in the extensions of youth unemployment. Interestingly, this stability breaks for the top three richer regions, which show again radical transitions.

In the French case, there is a great stability of the extensions of youth unemployment until a drastic change occurs when approaching the top 20% more dense regions, which reveals differences in how the labour market behaves in areas with different social setups.

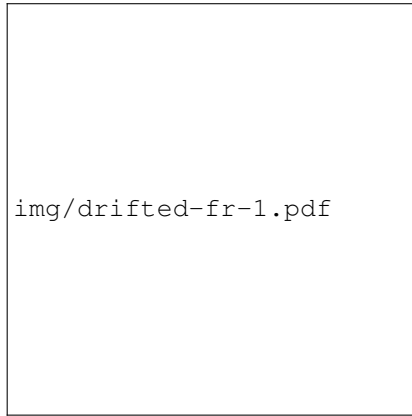
The French results confirm the tendency of the Australian case: approaching the so-called *developed areas* has a strong impact on economic indicators like youth unemployment. It also reveals the intrinsic socioeconomic relationship between the two di-



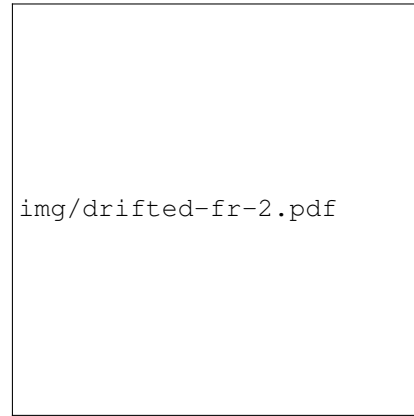
(a) Extensional drift per Australian region. P-values below horizontal line s denote drift. Regions by ascending GDP per capita.



(b) Evolution of relative distances d_k in Australian regions. A decrease in y-values denotes drift.



(c) Extensional drift per French *departement*. P-values below s denote drift. Regions by ascending population density.



(d) Evolution of relative distances d_k in French *departement*. A decrease in y-values denotes drift.

Figure 9.1: Plots of p-values and d_k distances for radical transitions in Australian and French regions for the concept *youth unemployment*. The x-axis represents regions in the dimensions GDP per capita (a, b) and population density (c, d) in ascendant order; the y-axis represents p-values (a, c) and relative drift distances d_k (b, d).

mension functions, given that in these countries the *GDP per capita* and the *population density* measure similar phenomena and, consequently, produce similar orderings.

9.6 Conclusions and Further Work

In this paper we present a generic framework for the study of radical transitions in concept extensions. Under the assumption that the data satisfy the typical setup of statistical datasets found in Linked Data and National Statistical Offices (NSOs), in this framework we formalize *tables* in canonical form as the basic structures where concepts and their extensions are contained. We establish basic functions for *rewrite* and *selection* to support basic table transformations, and we introduce how these tables can be ordered according to some chosen *dimension* of interest. When these dimensions satisfy the *smoothness assumption*, meaningful *radical transitions* can be found in the data by applying extensional concept drift detection techniques.

We follow these steps to apply the framework of radical transitions in two case-studies and multiple datasets. According to our results, we claim that the smoothness assumption is satisfied by the chosen dimensions of *time*, *GDP per capita* and *population density* for the study of radical transitions in the concepts *occupation* and *youth unemployment* in historical Dutch censuses and current censuses of Australia and France, respectively.

We contrast these results with knowledge of domain experts in the socio-historical and economical domains. We find radical transitions in 42 out of 217 (19.35%) historical occupational titles of the Dutch censuses between the years 1889 and 1899, concretely on occupations that were deeply affected by the changes of the late industrialization of the Netherlands. In the second case-study, we find radical transitions in 5 out of 8 (62.50%) Australian regions and 2 out of 96 (2.08%) French *departements* with respect to youth unemployment. Although the former shows a tendency on variation and the latter towards stability, it is noticeable that top-*k* regions are prone to radical transitions in both cases. Hence, we consider the chosen dimensions to satisfy the smoothness assumption, and their resulting orderings to be equally valid for the discovery of meaningful radical transitions in concept extensions.

We plan further work depending on to what extent our assumptions on the data hold. First, given that there no notion of identity between concepts in different tables is provided beforehand (which we suppose provided by experts in this work), we will extend the framework with the study of label drift and intension drift to track identity. Moreover, we will investigate how different implementations for label and intension drift affect the detection of radical transitions. We also plan to use other statistical tests to evaluate change in concept extensions, compare their results with Wilcoxon's, and study the differences between them. Supposing that the assumption on aggregated data does not hold, and therefore that individual observations are available, we will investigate how instance matching techniques can help in the problem of detecting extensional change of meaning. Finally, we plan to perform a systematic classification on how different dimensions behave in other datasets and various setups of an extended framework for radical transitions in concept extensions, intensions and labels.

Chapter 10

Predicting Concept Drift in Linked Data

10.1 Introduction

Motivation. Knowledge Organization Systems (KOS), such as SKOS taxonomies and OWL ontologies, play a crucial role in the Semantic Web. They are at the core of any Linked Data vocabulary and provide structured access to data, formalize the semantics of multiple domains, and extend interoperability across the Web. *Concepts* are central entities in KOS and represent objects with common characteristics. However, with time, objects are continuously subject to change. As the world changes, our understanding of it evolves. Concepts in KOS are inherently affected by *concept drift*, the change of their meaning over time [166].

What is the problem? Curation is a manual, knowledge intensive and arduous task. As domain concepts drift their meaning, curators and maintainers release different dataset *versions* updating their KOS, mostly using their expert knowledge. Concept drift positions them in a great dilemma: the continuous evolution of knowledge has a severe impact on their update work. Automatic detection of concept drift in their curation process would be a great aid. Unfortunately, there is currently hardly any tool to help dealing with concept drift.

Use-cases. To enhance comparability studies in the history of work, social scientists and historians have developed the Historical International Standard Classification of Occupations (HISCO), a taxonomy of tens of thousands of occupations from countries and languages around the world since the 16th century¹. Many historical datasets update their mappings to HISCO across their different versions. The Gene Ontology (GO) standardizes the representation of gene attributes across species and datasets. A new version is released every month.² Wikipedia categories and the DBpedia ontology are

¹See <http://historyofwork.iisg.nl/>

²See <http://www.geneontology.org/>

updated to improve cross-domain knowledge access. Librarians, social scientists, historians or biologists have a dire need to assess this versioning processes by identifying, and predicting, when a concept, class or code will change in the forthcoming release. This will reduce time spent in manual data exploration or requirements gathering.

Contribution. This paper introduces a generic approach to predict concept drift. Previous approaches have proven to be effective in predicting ontology *extension* in the *biomedical domain* [137]. We take this work as a basis to develop a *domain independent concept drift detection framework* based on supervised learning on past versions of Linked Datasets. We build a generic tool to aid data curators in maintaining datasets and increasing their proactivity towards change over time. This is useful in two scenarios: *refinement*, i.e. helping curators to revise a past version to check its coherence; and *prediction*, i.e. helping curators to be aware of concept drifts to come, allowing them to give higher maintenance priority to concepts that are more likely to change than others. We show viability of our approach in social sciences, history, and encyclopedic and cross-domain knowledge, adding to the already investigated biosciences [137].

Research Questions. In order to show validity of our approach we focus our research on the following three concrete research questions:

- **RQ1.** Can past knowledge be used to predict concept drift? Can this be done by extending an extension prediction method ([137]) into a concept drift prediction method?
- **RQ2.** What features encoding past knowledge have a greater influence on future drifts? What classifier performs best to predict these drifts?
- **RQ3.** Can this new method predict concept drift in a domain-independent manner? Can our algorithms predict drift on any Linked Data dataset?

Findings. In order to answer the research questions we run two refinement experiments in the Dutch historical censuses and the DBpedia ontology datasets, and a prediction experiment in the DBpedia ontology dataset. We obtain solid refinement/prediction performances, with f-measures of 0.84, 0.93 and 0.79 on test data in these experiments. We foresee a useful and effective domain-independent concept drift detection tool to aid data curators in their versioning processes.

The rest of the paper is structured as follows. In Sections 10.2 and 10.3 we survey previous efforts to address change, and define our target problem and formalism. Section 10.4 describes our approach, pipeline and feature set. In Section 10.5 we perform an experimental evaluation, describing the input data, process and results. In 10.6 we discuss these results and their relation with our research questions, before we conclude in Section 10.7.

10.2 Related Work

Concept drift is a very active research topic in Machine Learning. It is difficult to learn in real-world domains when “the concept of interest may depend on some *hidden context*, not given explicitly in the form of predictive features. (...) Changes in the

hidden context can induce more or less radical changes in the target concept, which is generally known as *concept drift* [154]. Hence, drift occurs in a concept when the statistical properties of a target variable (the *concept*) change over time in unforeseen ways. Multiple concept drift detection methods have been developed [95].

With the advent of the Semantic Web, changes in concepts have been investigated by formally studying the differences between ontologies in Description Logics [99]. Fanizzi et al. [92] propose a method based on clustering similar instances to detect drifts. Wang et al. [166] define what concept drift is and how to identify it in a Semantic Web setting. The related field of *ontology evolution* deals with “the timely adaptation of an ontology and consistent propagation of changes to dependent artifacts” [64]. As stated by Stojanovic [150], the first step for any evolution process consists in identifying the need for change; change capturing can then be studied as *struture-driven*, *data-driven* or *usage-driven*. Accordingly, *change* is only a step in the evolution process, although the definition of the goal of *ontology change* (“deciding the modifications to perform upon an ontology in response to a certain need for change as well as the implementation of these modifications and the management of their effects in depending data, services, applications, agents or other elements” [?, 116, 102]) suggests that the overlap between the two fields is considerable. Pesquita and Couto [137] develop the closest match to our work. They propose a method based on supervised learning on past ontology versions to predict extension of biomedical ontologies, using Stojanovic’s guidelines to design good predictors of change.

The need of tracing concept drift in application areas of the Semantic Web has been stressed, particularly in the Digital Humanities [124] and Linked Statistical Data, where concept comparability [83] and extensional drift [128] are key.

10.3 Problem Definition

10.3.1 Concept Drift

We use the framework of concept drift proposed by Wang et al. [166] in order to provide a definition and formalism.

The *meaning of a concept* C is a triple $(\text{label}(C), \text{int}(C), \text{ext}(C))$, where $\text{label}(C)$ is a string, $\text{int}(C)$ a set of properties (the *intension* of C), and $\text{ext}(C)$ a subset of the universe (the *extension* of C).

For the specific goal of this paper, we define the intension of a concept $\text{int}(C)$ as the set of structural relationships of C with other concepts, indicated by a set of user-defined predicates (e.g. `skos:broader`, `rdfs:subClassOf`). We define the extension of a concept $\text{ext}(C)$ as the set-members of C . We define the label of a concept $\text{label}(C)$ as the set of labels given to C .

All the elements of the meaning of a concept can change. To address concept identity over time, authors in [166] assume that the intension of a concept C is the disjoint union of a rigid and a non-rigid set of properties (i.e. $(\text{int}_r(C) \cup \text{int}_{nr}(C))$). Then, a concept is uniquely identified by some essential properties that do not change. The

notion of identity allows the comparison of two variants of a concept at different points in time, even if a change on its meaning occurs. For the specific goal of this paper, we assume that concepts with identical rigid intensions will have the same URIs³.

Two concepts C_1 and C_2 are considered identical if and only if, their rigid intensions are equivalent, *i.e.*, $int_r(C_1) = int_r(C_2)$.

If two variants of a concept at two different times have the same meaning, there is no concept drift. Intensional, extensional, and label similarity functions sim_{int} , sim_{ext} , sim_{label} need to be defined to quantify meaning similarity. These functions have range $[0, 1]$, and a similarity value of 1 indicates equality. For the specific goal of this paper, we define $sim_{int}(C', C'')$ as the function that returns whether C' and C'' have the same direct structural features. We define $sim_{ext}(C', C'')$ as the function that returns whether C' and C'' have exactly the same cardinality or not. We define $sim_{label}(C', C'')$ as the function that returns the minimum edit distance between the set of all labels assigned to C' and the set of all labels assigned to C'' .

We implement this framework as our definition of concept drift, and we instantiate it by concretizing the mentioned sets and functions for the specific goal of this paper. Our approach aims at genericity and these choices, as well as the overall drift definition, are customizable by the user (see Section 10.4).

A concept has extensionally drifted in two of its variants C' and C'' , if and only if, $sim_{ext}(C', C'') \neq 1$. Intensional and label drift are defined similarly.

10.3.2 Usage

We study concept drift in two different scenarios, depending on whether the versioned Linked Dataset is *closed* or *open*. We define a *closed dataset* as the dataset which versions were released in the past and their production finished at a certain point; consequently, no new versions will come. We define an *open dataset* as the dataset which updated versions are currently under production; new versions are expected to come.

Refinement. Given that no future versions will come, the prediction of concept drift in closed datasets is meaningless. In this scenario, we propose *refinement*. The purpose of refinement is to check the coherence of an intermediate version (instead of a version to come) with respect to previous versions.

Prediction. In open datasets, the primary question is to predict concept drift in a next version. To address this we propose the *prediction* usage. In prediction we use all available past versions to train a classifier and predict which concepts will drift in a next release. Obviously, *refinement* can also be applied to open datasets to check their internal coherency.

³We are aware that, in the general context of the Semantic Web, this assumption is not true: different versions of concepts should get distinct URIs, while different URIs may point to exactly the same concept due to the distributed nature of the Web. In general, users need to define their own identity functions between concept versions, *e.g.* using `owl:sameAs` or `skos:exactMatch` mappings.

10.4 Approach

The basic assumption of our proposed approach is that the knowledge encoded in past versions of a Linked Dataset can be used to faithfully predict which parts of it will suffer concept drift in a forthcoming version. This idea is inspired by the work by Pesquita and Couto [137] on predicting the extension of biomedical ontologies. Their approach derives from change capturing strategies that are based on implicit requirements. Traditionally, these requirements are defined manually on an expert knowledge basis. Their work tries to learn these requirements from previous extension events by selecting features that, according to Stojanovic [150], may have an influence in changing parts of an ontology:

- **Structure-driven**, derived from the structure of the ontology (e.g. if a class has a single subclass, both should be merged).
- **Data-driven**, derived from the instances that belong to the ontology (e.g. if a class has many instances, the class should be splitted).
- **Usage-driven**, derived from the usage patterns of the ontology in the system it feeds (e.g. remove a class that has not been accessed in a long time).

The state of the art [137] has proven success in the use of these features (i) to predict *extension*, that is, to estimate if a class will be extended (e.g. with new children or properties) in the future; (ii) in (OBO/OWL) ontologies; and (iii) in the biomedical domain (using the Gene Ontology). However, it remains unclear if supervised learning can be generally applied (I) to predict *drift*, that is, to estimate if a concept will experience change in its meaning; (II) in any Linked Dataset (i.e. generic RDF graphs); and (III) in a domain-independent manner.

Consequently, we extend the three essential aspects (i), (ii), (iii) of [137] in order to clarify the more generic (I), (II), (III) issues. These extensions, as we show in this Section, are not trivial. In the following we present a pipeline that includes: (a) an abstraction of the input parameters required for the learning process; (b) an abstraction of features that apply not only to OBO/OWL ontologies, but to any Linked Dataset; and (c) a pre-learning optimisation technique to merge features of identical versioned concepts into single training/test individuals. As a result, our approach tackles **RQ1** by providing generic and customizable change definition functions (including those in [166] (see Section 10.3.1)); generic and customizable features, including free choice of predicates/RDF types to use in their generation; customizable machine learning algorithms (feature selection and classification); and systematic and fully automated executions –from input Linked Dataset versions to output feature/classifier performances.

10.4.1 Pipeline

Figure 10.1 shows the pipeline of our proposed approach. The data flow starts in the upper left corner: the system gets the input set $\{\textit{Feature generation parameters, change definition, dataset versions, learning parameters}\}$, and returns the output set $\{\textit{Feature selection, classifier performance}\}$ in the bottom right corner.

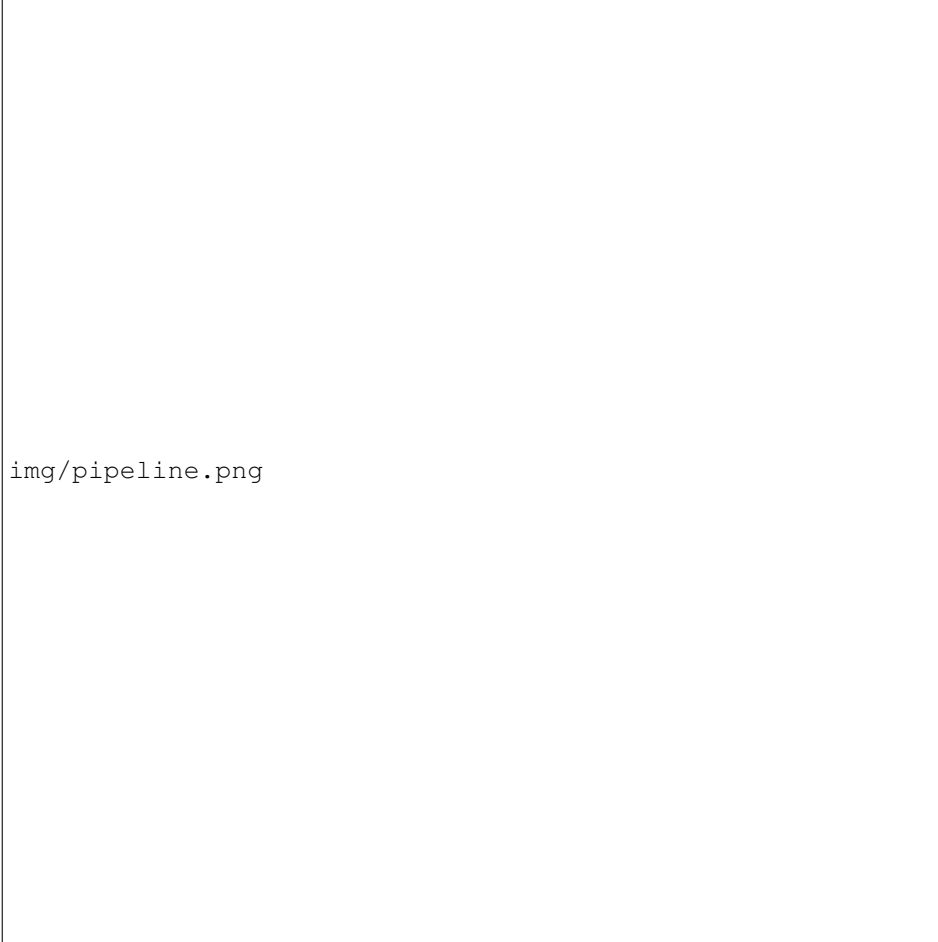


Figure 10.1: Pipeline of our approach. Arrows show the data flow through the modules.

The first module is the **Feature Generator (FG)**. It generates k training datasets and one test dataset, according to the following input set elements: (a) (N) *dataset versions*, in any RDF serialization, where the concept drift prediction is to be performed; (b) several user-set *feature generation parameters* that control the feature generation process (the $\Delta\mathbf{FC}$ parameter, setting the version to be used to decide if a concept of the training dataset has drifted or not; and the $\Delta\mathbf{TT}$ parameter, setting the version to be used to decide if a concept of the test dataset has drifted or not); and (c) a customizable *definition of change* that determines the value of the target variable. The last element of the input set, *learning parameters*, is passed further to be used in a later stage.

Once all set, k training datasets and the test dataset are built by the FG as shown in Figure 10.2. The parameters N , ΔFC and ΔTT are used to determine which versions will play the role of $\{V_t\}$, V_r and V_e . $\{V_t\}$ is the set of *training versions*, which are used to build the training dataset. V_r is the *reference version*, against which all versions in $\{V_t\}$ are compared, using the *definition of change* provided as input, to determine whether there is concept drift or not. V_e is the *evaluation version* and is used to build the

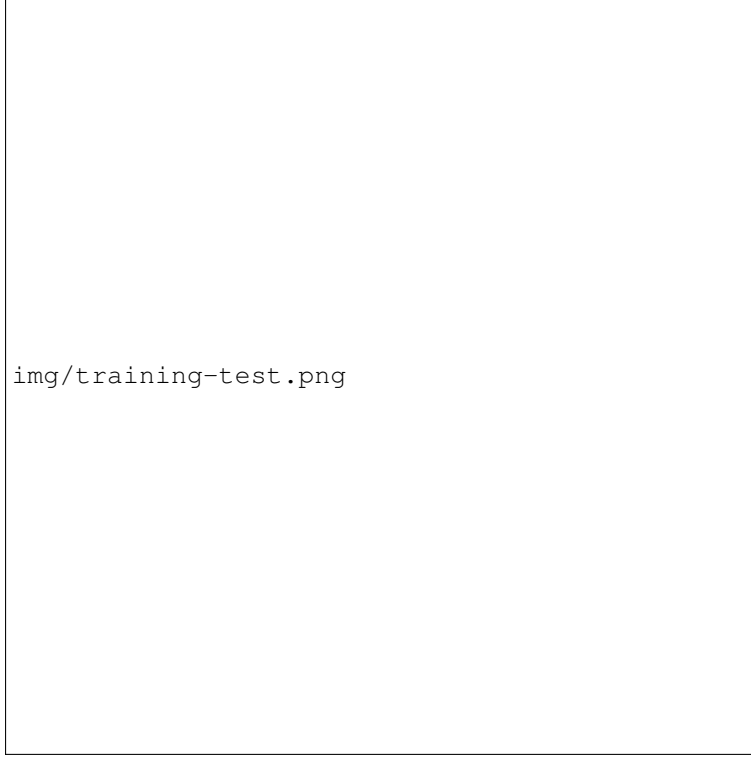


Figure 10.2: Building the training and testing datasets for $N = 7$, $\Delta FC = 1$ and $\Delta TT = 2$.

test dataset, following a similar procedure as with $\{V_t\}$ and V_r , this time comparing V_r with V_e . V_e is set by default to the most recent version. While extracting the features for each concept in the N versions, each exemplar is labeled depending on whether concept drift happened between one version of the concept and the next. This set of labels defines the target variable for the classification task. Our framework aims at genericity, and users can easily define their own notion of change and drift. We use the concept drift definitions described in Section 10.3.1 to decide if a concept drifts or not between two versions.

The FG produces k training datasets because versions can only be compared pair by pair. Although appending these k datasets into a single one is a possibility, it is important to preserve identity of the learning instances. Solving this is the purpose of the **Identity Aggregator (IA)**. The IA matches individual concepts of the k training datasets and merges their features into a single individual, modifying the dataset dimensionality accordingly. To do this, we match the URIs of the individual concepts; if matching concepts cannot be found, we fill features with NA values and modify the target variable value accordingly.

The single training and test datasets are then ingested by the **Normalizer (Norm)**. Norm adjusts value ranges, recodes feature names and types, and discards outliers, producing normalized versions of both datasets.

Finally, the training and test datasets are used by the **Machine Learning Interface**

(MLI) as an input for the feature selection and classification tasks. These are done in a generic and customizable way, building on top of the implementation of state-of-the-art machine learning algorithms contained in the WEKA API [103]. The last element of the pipeline’s input set, *learning parameters*, is used here to achieve this and contains: (a) the specific feature selection algorithm (from those included in WEKA) that ranks the features according to their relevance to predict drift; (b) a threshold t ; only features with a relevance higher than t will be selected; and (c) the list of classifiers to be trained. First, the module runs the chosen feature selection algorithm to select the most relevant features. Secondly, it trains systematically any subset of all WEKA classifiers included in the WEKA API, as chosen (by default all classifiers are trained). Lastly, it evaluates the trained models, and stores the results in disk.

10.4.2 Feature Set

We propose a set of features based on *structural properties* and *membership properties* of concepts as attributes for learning.

Structural features measure the location and the surrounding context of a concept in the dataset schema, such as children concepts, sibling concepts, height of a concept (i.e. distance to the leaves), etc. Since classification schemas are graphs in general and may contain cycles, these properties are defined with a *maxDepth* threshold that indicates the maximum level at which the property will be calculated (e.g. direct children, children at depth one, two, etc.). A concept is considered to be a child of another if they are connected by a user-specified property (e.g. `skos:broader`, `skos:narrower` or `rdfs:subClassOf`). We use *direct children* (descendants at distance 1) [*dirChildren*], *children at depth $\leq \text{maxDepth}$* [*dirChildrenD*], *direct parents* (concepts this concept descends from) [*parents*], and *siblings* (concepts that share parents with this concept).

Membership features measure to what extent a concept in the classification is used in the data. A data item in a Linked Dataset is considered to be using a concept of the classification if there is a user-defined membership property linking the data item with the concept (e.g. `dc:subject` or `rdf:type`). We use *members of this concept* [*dirArticles*] and *total members considering all children at depth $\leq \text{maxDepth}$* [*dirArticlesChildrenD*] as membership features. Finally, we define a set of hybrid features that combine the previous features into a single value (e.g. ratio of members per number of direct children) [*ratioArticlesChildren*, *ratioArticlesChildrenD*]. These sets of features map conveniently to the different types of change discovery described by Stojanovic [150]: structural features implement structure-driven change discovery; and membership features can be seen both as data-driven (since they describe instances belonging to the ontology) and usage-driven (since users querying these are indirectly using their classes).

These features are computed for each concept in all versions as indicated by the training and test dataset building parameters (see FG module, Section 10.4.1). However, not all of them may be used for predicting concept drift. [137] shows that similar features based on Stojanovic’s criteria [150] are good candidates. However, since it is

unclear whether or not these features explain well enough concept drift in arbitrary domains, we only use those that prove to be good predictors of concept drift in the feature selection phase (see MLI module, Section 10.4.1).

10.5 Evaluation

We apply our proposed approach to different versioned Linked Datasets⁴. We describe the properties of such datasets, the experiment setup and the evaluation criteria. We report on our results, providing evidence to **RQ2**: we evaluate (a) the performance of the feature set as a generic predictor of concept drift (see Section 10.4.2) and (b) the performance of the classifiers at the predicting task.

10.5.1 Input Data

In order to study the genericity of our approach and its applicability in a domain-independent setting, we use a set of multi- and interdisciplinary Linked Data datasets for which several versions have been published. Concretely, we use the 8 latest versions of the DBpedia ontology, and the last 8 versions of the Dutch historical censuses dataset (CEDAR)⁵. Each version consists of (a) a schema in SKOS, RDFS or OWL; (b) a set of data items making use of such schema; and (c) a set of labels describing the nodes of the schema.

In the case of the DBpedia ontology, the classification is the DBpedia OWL ontology itself [120], a community-curated formalization of all classes and properties describing DBpedia content. The data items are all resources of DBpedia which have some class of the ontology as `rdf:type`. The set of labels are the `rdfs:label` literals attached to the classes of each versioned ontology. In the case of the Dutch historical censuses (CEDAR) dataset, the classification is a SKOS hierarchy of HISCO occupations reported in each version. The data items are all census observations of people having one of these HISCO occupations as `cedar:occupation` (this is, the number of people employed in a certain occupation in a given version). The set of labels are the `skos:prefLabel` (Dutch) literals used in the census to describe these occupations in each specific version. A summary of the selected datasets is shown in Table 10.1.

The selection of these specific datasets is due to multiple reasons. First, they cover different levels of semantic expressivity, from SKOS taxonomies to OWL ontologies. Second, the temporal gap between each version varies from 10-12 months (in DBpedia) to 10 years (in CEDAR). Third, the selection contains both manually and automatically created datasets: the DBpedia ontologies have a mixed automatic/manual maintenance [120], while the CEDAR data is a totally manually maintained dataset. Fourth, DBpedia data is born-digital, open, and still evolving (2001–), whilst the CEDAR dataset is historical legacy, non born-digital, and temporally closed (1795–1971).

⁴See implementation details and extended results at <http://goo.gl/rASX6S>

⁵See <http://www.cedar-project.nl/>

Dataset	Version	N. concepts	N. relations	N. members
DBpedia ontology	(3.3) May 20th, 2009	282	174	3.8M
	(3.4) September 24th, 2009	334	204	4.3M
	(3.5) March 16th, 2010	255	255	5.2M
	(3.5.1) March 16th, 2010	257	257	5.5M
	(3.6) October 11th, 2010	272	272	6.2M
	(3.7) July 22nd, 2011	319	610	9.3M
	(3.8) June 1st, 2012	359	369	13.2M
	(3.9) April 3rd, 2013	529	541	15.9M
CEDAR	November 19th, 1849	408	405	76K
	December 30th, 1859	310	307	16K
	December 1st, 1869	238	236	7K
	December 31st, 1889	546	543	723K
	December 31st, 1899	566	563	837K
	December 31st, 1909	685	682	334K
	December 31st, 1920	271	268	103K
	December 31st, 1930	584	581	78K

Table 10.1: Summary of the selected datasets. *Version* refers to the date in which a specific version was published; *N. concepts* is a count of concepts/classes in that version; *N. relations* is a count of relationships connecting these concepts/classes; and *N. members* is the number of individuals making use of each classification.

10.5.2 Experimental Setup

Due to their inherent nature, CEDAR is a *closed* dataset, while DBpedia categories and DBpedia ontology are *open* datasets (see Section 10.3.2). For this reason, we divide our experimental setup in two parts: two *refinement* concept drift experiments using the CEDAR data (closed) and the DBpedia ontology (open); and one *prediction* concept drift experiment using the DBpedia ontology (open).

We set all parameters (see Section 10.4.1) as demanded by this setup. Since the datasets present temporal uniformity (see Table 10.1), we set $\Delta FC = \Delta TT = 1$.

Our evaluation process is two-fold. First, we quantify the quality of our features as concept drift predictors; we then choose the best performing features. We do this via *feature selection* (see Section 10.4.1). Second, we use this performant subset to evaluate the average quality of classifiers on predicting concept drift; we then choose the best performing classifiers.

To evaluate classifiers we follow a simple approach: we compare the predictions made by the classifiers with the actual concept drift going on in a next dataset version. To do this, we use the test dataset (see Section 10.4.1) produced after setting the parameter ΔTT . Since we compare predictions with unseen labeled data, we know whether the predictions are correct or not.

We evaluate the refinement and prediction experiments differently. Since the goal of refinement is to check coherence of the drift of concepts in an intermediate version, we execute several learning tasks adding more past versions to $\{V_i\}$ incrementally. We study how this impacts prediction of drift in V_i . The prediction experiments, on the other hand, consider all available versions, and we use the trained classifiers to predict concept drift in the most current version.

For assessing model quality, we use standard performance measures in machine learning: precision, recall, f-measure, and area under the ROC curve. Using these measures, we perform a two-fold evaluation. On one hand, we evaluate the quality of the models produced without making any predictions and using 10-fold cross-validation with the training data. In 10-fold cross-validation, the training set is divided in 10 equal folds:

Feature	Selection freq.	Feature	Selection freq.
<i>siblings</i>	10	<i>dirChildren</i>	21
<i>dirArticlesChildrenD2</i>	8	<i>siblings</i>	21
<i>ratioArticlesChildren</i>	8	<i>dirChildrenD2</i>	20
<i>dirArticles</i>	8	<i>dirChildrenD3</i>	20
<i>dirArticlesD1</i>	8	<i>dirChildrenD44</i>	20
<i>dirArticlesD2</i>	6	<i>dirArticlesChildrenD2</i>	17

(a) Top selected features for CEDAR.

(b) Top selected features for DBpedia.

Table 10.2: Top selected features in the CEDAR and DBpedia datasets.

9 are used to train the classifier, and the remaining one to test predictions (this is averaged on 10 repetitions in order to use all folds as test folds). On the other hand, we use the same indicators to evaluate the performance of prediction of the trained classifiers using the unseen test datasets V_e/V_i . In order to get insight on the prediction performance, we compare these measurements with a baseline implementing random prediction.

10.5.3 Results

Selected features for CEDAR and DBpedia

Table 10.2 shows the top selected features by the *Relief* algorithm [115], included in the WEKA API (see Section 10.4.1). The features are ordered according to the frequency in which they are selected across the versions. Table 10.2a shows that membership features (*dirArticles*, *dirArticlesChildren*) are systematically selected in the CEDAR data instead of structural properties (*siblings*, *dirChildren*). On the other hand, table 10.2b shows a clear preference for structural properties (*dirChildren*, *dirChildrenD*, *siblings*) in the DBpedia data, although some membership features have also an important relevance.

Refinement on CEDAR and DBpedia ontology

We perform a refinement (see Section 10.3.2) experiment using the CEDAR dataset. We execute our approach six times, adding one Linked Dataset version to $\{V_t\}$ and shifting V_i forward once each time. We identify each experiment with the year of the version to be refined $V_i \in [1869, 1889, 1899, 1909, 1920, 1930]$. Figure 10.3 compares the classification results for all runs.

We perform a refinement experiment with the DBpedia ontology versions. We execute our strategy six times over the eight versions, first taking the first three, and adding one to $\{V_t\}$ and shifting V_i once each time (see Section 10.3.2). Figure 10.4 shows the results, which can be compared with the results of the previous experiment shown on CEDAR data in Figure 10.3.

Prediction on DBpedia ontology

We perform a prediction (see Section 10.3.2) experiment on the DBpedia ontology. We execute our approach once, using all available versions as training set $\{V_t\}$, and leaving the last for testing (V_e). Table 10.3 shows the performance of classification.

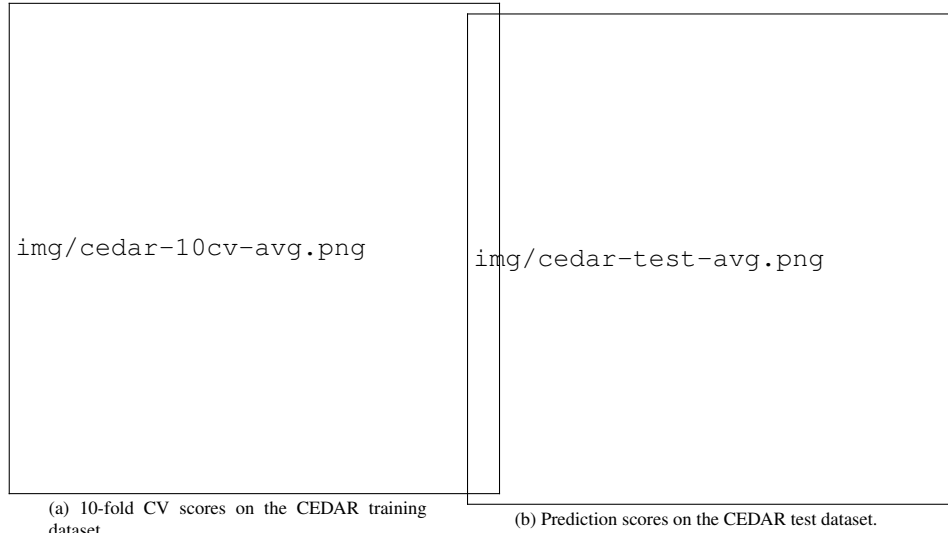


Figure 10.3: Average classifier performance in the CEDAR refinement experiment with 6 incremental learning runs. Lines show performance measures varying along them.

	$V_e = 2013$
Precision	.98
Recall	.98
F-measure	.98
ROC area	.81
Base precision	.48

(a) DBpedia ontology 10-fold CV scores, training dataset.

	$V_e = 2013$
Precision	.66
Recall	.75
F-measure	.67
ROC area	.58
Base precision	.52

(b) DBpedia ontology prediction scores, test dataset.

Table 10.3: Average classifier performance of prediction in the DBpedia ontology.

10.6 Discussion and Lessons Learnt

Table 10.2 shows that selection of features of one kind or another is influenced by the nature of the dataset under consideration. On the one hand, membership features are ranked as the most important ones to predict concept drift in CEDAR (i.e. knowledge on how instances are linked to the schema). On the other hand, structural features are top ranked to predict drift in DBpedia (i.e. knowledge about the basic structure of the ontology). Both kinds of features are selected for both datasets at some point, though.

Figures 10.3 and 10.4 show that classification outperforms the simple random baseline. Although the Logistic, the MultilayerPerceptron and the tree-based algorithms have good performance in specific situations, the NaiveBayes classifier shows consistent results in all concept drift refinement and prediction experiments. Similar behavior and results have been described in [137]. Interestingly, we observe how the non-overfitting tendency of NaiveBayes is an advantage if the classifier is trained with more past versions. MultilayerPerceptron, for instance, predicts better with less data (f-measures from 0.82 to 0.30), but with more versions NaiveBayes wins (0.72 to 0.84) (see Figure 10.3).

In the refinement experiments performance is affected by two factors: the amount of

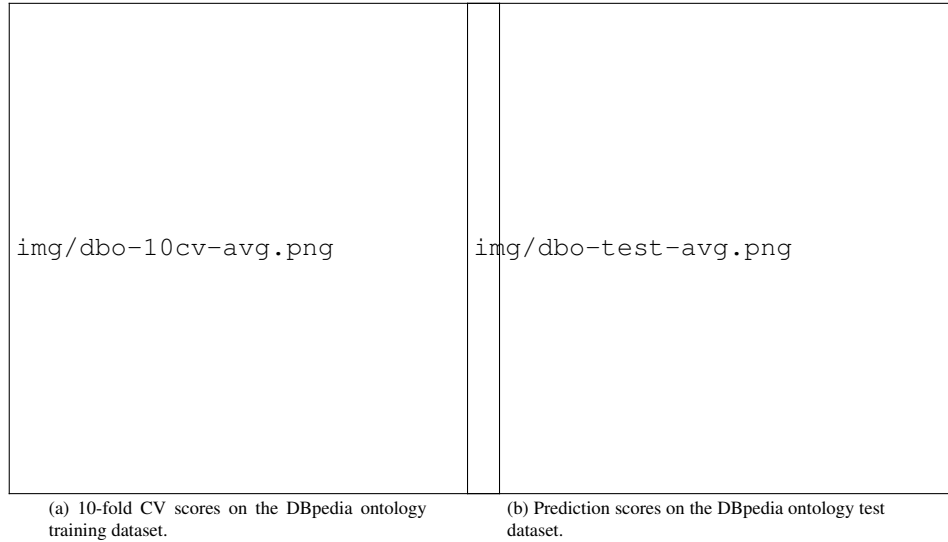


Figure 10.4: Average classifier performance in the DBpedia ontology refinement experiment with 6 incremental learning runs. Lines show performance measures varying along them.

past versions used to learn, and the characteristics of these versions versus the version to predict. In general we observe more performant models at predicting V_i when more versions are added to $\{V_t\}$. However, CEDAR’s 1889 and 1930 versions DBpedia ontology’s 3.5.1 and 3.7 are clear exceptions (see Figures 10.3b and 10.4b). This can be due to several reasons. First, the version V_i to be predicted may contain unexpected changes that have not been learned from previous versions. CEDAR versions of 1889 and 1930, for instance, are known to have suffered a major restructuring or revision almost from scratch [119], making their drifts harder to predict. Second, corner cases of concept drift might not be captured with the feature set. Third, these CEDAR versions contain scarce member data (see Table 10.1), and lack of data about uncommon types of drift seems to hold a relationship on why prediction is harder in these cases. Still, our refinement approach proves to be useful on detecting these coherence data-issues.

Results also confirm that forthcoming concept drifts depend on past drifts, and that incorporating past knowledge about drift helps on predicting it. However, abusing this has a side effect, since adding too many old versions to the learning set $\{V_t\}$ seems to lower the prediction performance. This suggests that the inclusion of a moving window over $\{V_t\}$ could be a sensible solution.

<http://cedar.example.org/ns#hisco-06> is an example of a CEDAR/HISCO concept predicted to drift which in fact did: the class of “medical, dental, veterinary and related workers”. Most of its features present high stability across the versions; except those related to its members, i.e. the sets of people reported to belong to this concept. These vary from 841 sets of observations, to 68, 143, 662 and 110, while structural properties like number of children (4) or siblings (9) remain relatively stable. <http://dbpedia.org/ontology/CollegeCoach> is an interesting DBpedia concept also expected to drift. The number of Wikipedia articles pointing to it increases linearly (2787, 3520, 4036, 4870...); it always remains a leaf with a unique parent,

so its children subhierarchy does not change either. Interestingly, its siblings remain stable (21, 21, 23, 23) until it gets a new parent and its siblings suddenly explode (23, 344). Considering these observations, it is easy to see why membership and structural features are highly ranked for CEDAR and DBpedia, respectively.

According to this, an important lesson to be learnt is that *the degree of dependency between the changed parts of a dataset and well-known change-aware features can save enormous amounts of update work*. We have shown that these features are collectable, rankable and usable in a fully-automatic and customizable pipeline to predict future drifts in Linked Data. If this degree of dependency is high, knowledge engineering tasks can be automated (by the classifier) to avoid coherency mistakes or taking too narrow decisions.

10.7 Conclusions and Future Work

Concept drift poses enormous challenges to data curators and maintainers. For those publishing their data as Linked Data, change of meaning of concepts implies the continuous release of versions, which is a knowledge-based and labor-intensive task. Our approach detects automatically which parts of a Linked Dataset will likely undergo concept drift in a forthcoming version, using supervised learning and leveraging drift knowledge contained in past versions.

Recalling back our research questions and results, the assumption that concept drift can be predicted in refinement/prediction scenarios using past knowledge is acceptable considering intensional, extensional and label drifts. We have shown how to achieve this by generalizing the state of the art in a Machine Learning for Linked Data pipeline (**RQ1**, see Section 10.4). We have studied the variance in relevant features from our feature set, and how classifiers behave using these features to predict concept drift (**RQ2**, see Section 10.5). With respect to its domain-independent applicability (**RQ3**), we predict drift accurately using non-domain related features in sociohistorical and cross-domain datasets, which add to the list of biomedical ontologies [137]. We get encouraging prediction results, obtaining an f-measure of 0.84, 0.93 0.79 for our selected classifier when predicting unseen data using sociohistorical, encyclopedic and cross-domain datasets. We study the impact of the addition of more past knowledge, and the limitations of the approach regarding from-scratch restructuring of datasets.

Prediction of concept drift may pose new opportunities for data publishers, increasing the efficiency of their curation pipelines, reducing curation time and allowing them to prioritize maintenance on drifting concepts. Additionally, it can be used to quantify the stability of a dataset or, alternatively, the “unexpected” changes across versions. This is key for the fundamental issue of *preservation of Linked Data*, which aims at archiving of Linked Data and suggests that stable data should be archived first. Our method can be used to measure such stability.

Multiple challenges are open for the future, and we plan to extend this work in several aspects. First, how alternative definitions of concept drift affect the prediction of further drifts poses an interesting question. Second, we plan to apply our approach in

other domains to continue investigating its genericity. Third, we intend to use a higher number of versions in order to confirm the trends we observed and to investigate optimal window sizes. Finally, we envisage an extension of our approach in a big data environment, in order to cope with larger datasets and detect change of meaning of concepts in real time.

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